

CAPSTONE PROJECT 1
Proposal

Gamified Mobile App for Student Procrastination Management

by

KEERTANA A/P SUBRAMANIAM
23109614

Bachelor of Information Systems (Honours) (Data Analytics)

Supervisor: Associate Professor Ts. Dr Lee Yun Li

Semester: September 2025

Date: 2 January, 2026

School of Computing and Artificial Intelligence
Faculty of Engineering and Technology
Sunway University

Contents

Abstract	vi
1.0 Introduction.....	1
1.1 Background	1
1.2 Problem Statement	3
1.3 Aim of the Project	4
1.4 Objectives of the Project	4
1.5 Project Scope and Methodology	5
1.6 Proposed Timeline and Expected Outcomes	6
1.7 Research Structure.....	7
2.0 Literature Review.....	9
2.1 Introduction to Literature Review	9
2.2 Understanding Academic Procrastination	9
2.2.1 Definition and Concepts	9
2.2.2 Factors Influencing Procrastination.....	10
2.3 Machine Learning in Academic Procrastination	12
2.3.1 Introduction	12
2.3.2 Statistical and Structural Modelling Approaches	13
2.3.3 Machine Learning for Behavioural Analysis and Change.....	15
2.3.4 Predictive Modelling of Procrastination.....	18
2.3.5 Limitations and Challenges	22
2.4 Gamification and Its Role in Behaviour Change	24
2.4.1 Concept of Gamification	24
2.4.2 Gamification for Procrastination	26
2.4.3 Limitations of the Gamified Approaches.....	28
2.5 AI-Based Hierarchical Task Decomposition and Planning	30
3.1 Overview	33

3.2 Process Flow	34
3.3 Dataset Description	35
3.4 Machine Learning Pipeline	36
3.4.1 Phase 1: Data Pre-processing	36
3.4.2 Phase 2: Feature Selection and Secondary Dataset Construction.....	37
3.4.3 Phase 3: Model Training and Evaluation.....	39
3.4.4 Application Development Pipeline	41
3.4.5 Phase 4: System Integration	45
3.4.6 Tools and Techniques Used	46
4.0 Work Plan.....	47
4.1 Project Deliverables	47
4.2 Capstone Project 1 - Research and Planning Phase	48
4.2.1 Work Activities and Deliverables for Capstone Project 1	48
4.3 Capstone Project 2: Implementation and Evaluation Phase.....	50
4.3.1 Work Activities and Deliverables for Capstone Project 2	50
4.4 Project Milestones with Risks and Mitigation	51
4.5 Gantt Chart.....	51
4.6 Chapter Summary.....	54
5.0 References.....	55

List of figures

Figure 1: Process Flow Diagram.....	34
Figure 2: Feature Importance Scores for the Top 10 Variables Selected from the Raw Dataset	38
Figure 3: Login page and new user registration interface.....	44
Figure 4: User registration form for creating a new account	44
Figure 5: Procrastination and academic behaviour survey used to assess user profile and risk factors.....	43
Figure 6: Main task page showing options to create tasks.....	43
Figure 7: Pop-up interface displayed when creating a new task from the main task page	43
Figure 8: Task page after a new task is created, including feedback alert for task creation	43
Figure 9: AI-powered task decomposition view showing subtasks	44
Figure 10: Points and rewards display indicating user progress after completing a task	44
Figure 11: Leaderboard page showing rankings and scores of other users within the app.....	44
Figure 12: Account settings page where users can manage profile and app preferences	44
Figure 13: Gantt Chart for Capstone Project 1	52
Figure 14: Gantt Chart for Capstone Project 2	53

List of Tables

Table 1: Summary of Personal, Emotional, and Cognitive Factors Influencing Procrastination Identified in Prior Studies	12
Table 2: Comparison of Machine Learning Techniques Used for Behavioural Analysis and Behavioural Change in Education	18
Table 3: Summary of Machine Learning Models and Performance Metrics Used in Prior Studies on Predictive Modelling of Academic Procrastination	21
Table 4: Summary of Related Studies on Predictive Modelling of Academic Procrastination and Their Identified Limitations	24
Table 5: Summary of Prior Studies on Gamification Elements Applied to Procrastination and Productivity Management.....	28
Table 6: Key Limitations Identified in Prior Studies on Gamification-Based Procrastination and Productivity Interventions.....	30
Table 7: Summary of Task Decomposition Approaches and Their Key Contributions	32
Table 8: Summary of Dataset Variables and Their Descriptions	36
Table 9: Top 10 Features Selected from the Raw Dataset Based on Importance Scores.....	38
Table 10: Final Variables Selected for the Secondary Dataset with Roles and Descriptions ..	39
Table 11: Project deliverables with durations and scheduled weeks	47
Table 12: Capstone Project 1 work activities, deliverables, durations, risks, and mitigation strategies	48
Table 13: Capstone Project 2 work activities, deliverables, durations, risks, and mitigation strategies	50
Table 14: Key project milestones with associated risks and mitigation strategies for proactive management	51

Abstract

Academic procrastination is a common problem among university. It negatively affects completion of task, learning quality, and self-regulated learning, despite significant research and existing time management and motivation strategies. Existing solutions use fixed strategies or rely on self-assessments that lacks personalization and timely support for change in behaviour. Hence, this project aims to build a gamified mobile app that uses machine learning to offer tailored procrastination support. A structured literature review will highlight the key factors that affects procrastination such as maladaptive perfectionism, self-efficacy, self-regulated learning, and resilience. The review also studies the current use of machine learning and gamification in education. An existing dataset on academic procrastination will help create a predictive modelling pipeline. A conceptual system design will combine prediction model with personalized nudges and task breakdowns. The goal of Capstone Project 1 is to lay a groundwork for methodology and design. This includes the project scope, system design, rationale of dataset, and a plan for development. These will help implement and assess the proposed system in Capstone Project 2.

1.0 Introduction

1.1 Background

Academic procrastination can be defined as the intentional and unnecessary delay of academic tasks. This happens even when students expect that waiting to complete the tasks will lead to negative consequences. This issue is common among students and consistently related to poorer academic performance, increased stress, and lower psychological well-being. Yan and Zhang (2022) made a thorough analysis and showed that procrastination has been studied extensively over the last three decades, especially focusing on the population of students. The study mainly revolves around defining what procrastination is, what causes it, and the effects of it, particularly in educational settings. Conscientiousness and self-regulation which are one of the factors related to personality along with emotional and cognitive processes often appears in the literature. Despite the research done that is ranged widely, Yan and Zhang (2022) pointed out that much of the existing work relies on self-reported, one-time data. This emphasizes the need for more methods that are systematic and predictive to understand the behaviour of procrastination.

Procrastination is heavily influenced by differences in traits of personality and everyone's emotional regulation in academic settings. A study by Cheng et al. (2023) shows that conscientiousness can significantly affects the tendency of students to procrastinate in online learning environments. Proactive and inhibitory aspects of conscientiousness were identified in the study. It showed that lower levels of conscientiousness are indirectly linked to higher procrastination through negative feelings related to achievement. Negative emotions like anxiety and frustration especially have the strongest relation to procrastination. This demonstrates that emotional experiences can affect personality trait by having a behaviour that delays tasks. The study also mentions that with limited experience with online learning environments, it can worsen these emotional issues, and that would lead to increase in procrastination. This indicates that poor time management is not the only factor affecting procrastination. Procrastination is a complex behaviour that is shaped by responses that are emotional to learning demands and perceived task difficulty.

Besides the traits of an individual, factors like contextual sensors can also significantly be the reason for procrastination behaviour. Sirois (2023) claims that stressful academic environments can make the students even more vulnerable to procrastination as it can drain their emotional and cognitive coping resources. In situations that are highly stressful, procrastination in terms

of regulation of mood acts as a low-effort way to avoid painful task-related emotions. The relationship between stress and procrastination can be dynamic as procrastination can increase stress. This can create a cycle of delay and emotional discomfort. These pressures are even stronger in today's educational settings. This is due to students facing more online, self-paced, and demanding learning environments. Overall, these studies prove that procrastination occurs due to interplay of personality traits, emotional regulation, and contextual stress. This illustrates the need for systems that are smart enough to identify procrastination risk early and help the students to manage their academic behaviour better.

Although there are various approaches that have been introduced to address the issue of academic procrastination such as time management techniques, self-regulated learning strategies, behavioural interventions. Additionally, more recently, digital tools like apps for productivity and gamified learning systems are also few of the approaches introduced to tackle procrastination. Research on self-regulated learning that are supported by technology shows that structured digital support can help students manage study activities and stay engaged. However, these systems still lack as it is still developing and often in short of personalized features and adaptive guidance (Pengel et al., 2021).

Self-regulated learning and motivation are linked to the outcomes of procrastination. For instance, one of the most important parts of self-regulated learning are intrinsic motivation for learning and academic self-efficacy. These factors relate to procrastination negatively among students. This leads to emphasizing the need for internal motivation and supportive educational environments (Chen et al., 2025). Meanwhile, predictive modelling techniques are becoming more popular in academics. This is because machine learning-based predictive models have shown a great capability to not only forecast the performance of students but also identify at-risk learners. This is done by handling educational datasets that are large and complex. By doing so, it demonstrates the potential for anticipatory intervention and personalized support (Almalawi et al., 2024).

Digital behaviour changes support systems (BCSSs) have become more common. An example of this system is that it can be used for developing mobile apps that help users address the issue of procrastination by using motivational design elements. However, since the research in this area is still new, it lacks thorough evaluation of their effectiveness and personalization features (Kirchner-Krath et al., 2025). Although these approaches have the potential, due to deep

personalization, adaptive task guidance, or effective integration of behavioural prediction with motivational support. As a result, procrastination remains a major problem for the students.

1.2 Problem Statement

Academic procrastination has been a major issue for a very long time. Since, modern learning environments require high levels of self-regulation and independence, it is even more evident. Procrastination is one of the most studied maladaptive academic behaviours as noted by Yan and Zhang (2022). However, it remains common among students across various disciplines and educational levels. Despite extensive theoretical study, procrastination continues to disrupt the students from completing their academic tasks on time. There is evidence from research that proves procrastination is related to delay of submission of task, heavy reliance on completing work last-minute, and increase in academic stress, which collectively weaken students' learning processes and outcomes. Students who procrastinate will demonstrate lower academic self-efficacy and decrease in intrinsic learning motivation. This will result in minimal engagement with learning materials and poorer learning quality because of time constraint for revision and reflection (Chen et al., 2025). This problem was made worse when the use of online and self-paced learning formats were increased. Even though students have more flexibility, they also have fewer structures to help regulate their study habits. As a result, procrastination has become a consistent challenge that negatively impacts the success in academics and learning sustainability rather than being an occasional lapse in behaviour.

The effects of academic procrastination extend beyond lower academic performance by involving significant psychological and emotional costs. Sirois (2023) stated that there are strong links between procrastination, increased stress, anxiety, and negative feelings. Procrastination often occurs to avoid unpleasant feelings related to tasks, especially when under stress from a mood regulation perspective. However, by doing it will lead to a greater long-term stress since it will create a cycle of emotional discomfort and delay. Cheng et al. (2023) further illustrate that negative emotions tied to achievements are some of the strongest predictors of procrastination in online learning environments. This is especially for students with lower conscientiousness or little experience with online learning. Hence, these studies concluded that procrastination is not just about poor discipline or time management. Rather, it is a complex response that is influenced by emotional vulnerability and contextual pressures.

Despite the impacts of procrastination that are known widely, existing interventions largely respond reactively and are often too general. Yan and Zhang (2022) note that most of the research on procrastination are based on data collection that are self-reported at single points in time. Thus, limiting the ability to detect behavioural risks dynamically or forecast future procrastination. Moreover, support methods that are traditional like academic counselling or general tips on time management often overlook the difference of the individuals in emotional triggers and behavioural patterns. Through these, there is a clear need for predictive and data-driven approaches that can identify students at risk of procrastination early and provide timely and personalized support. By filling these gaps, procrastination will be always consistently checked upon, hence reducing academic stress, increased engagement, and ultimately enhancing the learning outcomes of the students.

1.3 Aim of the Project

The aim of this project is to develop and evaluate a machine-learning based predictive model to identify students who are at risk of academic procrastination and then integrate this predictive framework within a gamified mobile application that delivers adaptive behavioural support. The proposed predictive framework will be able to detect procrastination at risk at an early stage and will provide timely intervention by leveraging both behavioural and psychological indicators. By integrating the machine-learning driven predictive modelling with the gamification mechanisms, the aim of this app is to support self-regulated learning. This will be done by encouraging sustained engagement, improving behaviours on managing tasks, and mitigating the negative academic and emotional consequences that is related to procrastination.

1.4 Objectives of the Project

The following objectives are defined so that this project can achieve the aim that is stated in the previous section:

1. To investigate procrastination behaviours and gamification strategies by reviewing existing studies on academic procrastination and motivational design to identify key behavioural factors and effective gamification approaches.

2. To develop a machine learning model for predicting procrastination and personalized interventions by selecting algorithms that are suitable, training predictive models on behavioural data, and assessing their performance in identifying procrastination risk.
3. To design a gamified mobile application that supports self-regulated learning by applying predictive insights with gamified elements to encourage sustained engagement and improved task management among students.

1.5 Project Scope and Methodology

This project focuses on developing a machine learning-based system to predict the procrastination risk of students and provide adaptive support through a gamified mobile application that will be developed using Flutter, which supports both Android and iOS platforms. It specifically targets university students and highlights understanding procrastination as a behaviour rather than a clinical issue. The goal of this project and this app is to help with self-directed learning by offering timely and personalized feedback and nudges. Besides that, the goal is also to provide structured support instead of therapeutic treatment or diagnoses.

The study uses a secondary dataset that is derived from a validated academic procrastination questionnaire by Rusdi et al. (2020). The methodology includes the process of how the secondary dataset was derived. It started from understanding the data, preprocessing it, selecting the features, and transforming it to create a dataset suitable for machine learning. Behavioural indicators of procrastination were given more attention to make sure that the predictions can lead to actionable outcomes within the app. The predictive challenge is framed as a supervised classification problem. Hence, making it easier to interpret risk levels and activate support mechanisms.

The methodological framework has two parallel pipelines which includes a machine learning pipeline and application development pipeline. The first pipeline which is machine learning is where model selection, training and validation, and evaluation using recognized performance metrics and a methodical data partitioning strategy will be included. The machine learning models that will be utilized are Logistic Regression, Support Vector Machine (SVM) Decision Tree, Random Forest, Gradient Boosting, k-nearest Neighbours (k-NN) and Multilayer Perceptron (MLP). These models will be evaluated using accuracy, precision, recall, F1-score,

and confusion matrix metrics. Simultaneously, the second pipeline which is the application development path covers the requirement analysis, gamification design, frontend development, backend logic and database management. This application prototype will be developed using the Flutter platform.

Furthermore, the application will also include an AI-assisted task structuring feature that breaks down user-centered academic tasks into smaller subtasks based on the description given by the user and will also provide internal deadlines for the subtasks. This feature uses rule-based and heuristic planning logic so that it can enhance task clarity and workload management. This helps the role of predictive modelling to recognize the risk of procrastination. The system is built for modular integration, which is why it allows the machine learning model and app components to connect through APIs. This ensures scalability and future improvements during later development stages.

1.6 Proposed Timeline and Expected Outcomes

This project is divided into two phases where the first phase is Capstone Project 1 (CP 1) and the second phase is Capstone Project 2 (CP 2). This is to ensure a clear manageable development process. There will be a total of 28 weeks to work on these phase where each phase will be allocated 14 weeks. CP 1 focuses on planning, theoretical grounding, and system design whereas CP 2 emphasizes on the implementation, integration and evaluation. By dividng the project into two phases, it allows enough time for analysis, justifying the methods used, and refining the project when necessary. By doing so, it can help ensure that the final prototype is both functional and meets user needs.

In the CP 1 phase, the project focuses on identifying the problem, conducting literature review and developing methodologies. Key activities includes reviewing existing research on academic procrastination, predicting procrastination risk using machine learning, and exploring gamification strategies. Besides that, this phase also defines the scope of the project, objectives, and system requirements. Furthermore, it involves designing the system architecture, addressing the machine learning path, gamified intervention logic, and overall workflow. By the end of CP 1, a complete project proposal and design specification will be ready to guide the development of the app in the next phase which is CP 2.

CP 2 phase is dedicated to implementing and evaluating the proposed system. This phase includes creating the trained machine learning model to predict the procrastination risk. Once trained, the best performing model will be integrated with the backend logic. Finally, a fully developed mobile application prototype featuring gamification will be ready. The AI-assisted task structuring component will also be created and linked to the workflow of the application. Tests on the model performance, system functionality, and basic usability will be conducted to evaluate the effectiveness of the predictive framework and how the application performs for users. Feedback from usability testing will help refine the system components and enhance overall reliability.

At the end of the project, the expected outcome is a completely developed prototype of a functioning gamified mobile application, and the application can work on multiple platforms such as Android and iOS. This app is integrated with a machine learning-based model for predicting procrastination risk. When the system detects the tendency of procrastination, the prototype is designed to provide personalized nudges, reminders and adaptive feedback. The system will also offer clear task guidance where it will break academic tasks into smaller, manageable subtasks and set internal deadlines. It will also offer guidance on tasks through gamified elements such as progress tracking, rewards, and reminders. Additionally, a completed project document on design choices, development processes, testing results, and evaluation findings will be compiled. Overall, this project aims to show how machine learning and gamification techniques can support self-regulated learning and reduce academic procrastination.

1.7 Research Structure

This research is divided into four chapters to guide the design and development of a machine learning-driven gamified mobile application to manage academic procrastination. The chapters are structured accordingly to ensure a logical flow from identifying the problem to planning the system and preparing for implementation. Chapter 1 of the research introduces the project, outlines the background of academic procrastination, problem statement, aim of the project, objectives, scope and methodology of the project, and proposed timeline. Chapter 2 presents a detailed literature review by analysing existing research on academic procrastination, machine learning methods for predicting behaviour, gamification strategies, and AI for breaking the tasks into subtasks. Chapter 3 describes the methodology for the project where it details the

machine learning framework and application development. Finally, Chapter 4 focuses on the work plan of the project, milestones using Gantt chart, and deliverables for both Capstone Project 1 and Capstone Project 2.

2.0 Literature Review

2.1 Introduction to Literature Review

This chapter provides a comprehensive overview of literature that is relevant to understanding academic procrastination and mitigation techniques. It examines the key concepts and theories from cognitive, emotional, and behavioural perspectives. The chapter also discusses how gamification can enhance motivation and engagement and how machine learning can be used to forecast student behaviour. By synthesizing these topics, the literature review highlights the current research gaps and lays the groundwork for the current study. This section seeks to better address academic procrastination by combining gamified interventions and predictive analytics.

2.2 Understanding Academic Procrastination

2.2.1 Definition and Concepts

Academic procrastination is a common behaviour influenced by factors that are emotional, cognitive, and situational. Despite knowing that it will have negative consequences, it is often seen as the delaying of tasks voluntarily. It plays a significant role as it affects students' academic performance and overall well-being. Research shows that procrastination involves issues that are deeply related to controlling emotions, motivation and habits other than just poor time management.

A study by Ahmed et al. (2023) looked into the prevalence of academic procrastination among university students and found high rates of behaviour of procrastination across various student tasks. These tasks include doing revision for exams and completing assignments. The common pattern among these students was the tendency of delaying tasks despite understanding it will affect their academic performance. This study highlights that procrastination is widespread and how it appears in different academic fields, proving that it is a serious problem among students.

Another significant study by Sirois (2023) proposed a model of procrastination to prove that stressful contexts such as academic pressure, personal stressor or the constant shift to digital learning can increase the risk for procrastination. This perspective views procrastination as a path for students to regulate their mood. This is where they avoid completing tasks to escape negative emotions rather than improve their time management.

Analysing these studies, Fentaw et al. (2022) examined how procrastination is shaped by emotional avoidance and behavioural patterns. The study found that students often delay completing their assignments when they feel overwhelmed, overburdened or unsure. The use delay as a temporary escape rather than planning a strategic decision.

Together, these studies show that academic procrastination involves emotional avoidance, recurrent behavioural cycles, and vulnerability in stressful situations. This understanding helps to explain how procrastination develops and persists among students. It also highlights the importance of creating intervention so that it can identify behavioural patterns, assist with emotional regulation, and adjust to students' actual learning environment.

2.2.2 Factors Influencing Procrastination

Procrastination among university students is influenced by personal characteristics, psychological processes and contextual factors. Research continuously shows that students' tendency to put off academic tasks is greatly influenced by maladaptive perfectionism, self-efficacy, self-regulated learning, and resilience. Designing solutions that are both successful and personalized requires an understanding of these aspects. Although there are multiple factors that are being examined by each study, this section focuses on the most influential factors that was recognized in each study. However, the full range of the examined factors is summarized in Table 1.

A study reveals that academic procrastination among Chinese nursing undergraduate students is strongly correlated with maladaptive perfectionism using a cross-sectional study among 587 undergraduate nursing in two undergraduate universities in China, and this effect is somewhat mitigated by students who have higher self-efficacy (Huang et al., 2023). However, self-efficacy was a partial mediator, which indicates that students with greater confidence in their capabilities were better equipped to deal with perfectionism. The relationship is further moderated by resilience as with greater resilience weakens the perfectionism-procrastination link. These results highlight how academic procrastination may be more likely in people with maladaptive perfectionism, poor resilience, and low self-efficacy. This demonstrates how crucial it is to support both emotional and cognitive resources while dealing with procrastination.

Another study investigated how self-regulated learning predicts procrastination and academic outcomes which was conducted on 728 students in Eastern Spain (García-Ros et al., 2023). The study used structural equation modelling (SEM) approach where the participants filled questionnaires on self-regulated learning, procrastination, academic stress, well-being, and achievement. Procrastination was proved to be acting as the mediator when two competing structural models were tested. Procrastination had direct negative effects on achievement and well-being and a positive effect on stress. It also partially had an effect on the impact of self-regulated learning on these outcomes. The study emphasizes that poor metacognitive strategies and low academic self-efficacy, which are the weaknesses in self-regulated learning, function as key factors that lead the students to engage in procrastination.

Jing et al. (2025) investigated how negative perfectionism contributes to academic procrastination among Chinese university students and whether self-efficacy acts as a mediating psychological mechanism in this association. A quantitative cross-sectional survey was designed to collect data from 460 students from Southwest China at four universities. Validated tools like Aitken Procrastination Inventory, an academic self-efficacy scale and the Negative Perfectionism Questionnaires were used for this study. It was predicted that higher the negative perfectionism the worse the academic self-efficacy is. This regression-based mediation analysis was done using Hayes' PROCESS Macro. The direct impact of perfectionism on procrastination became non-significant when self-efficacy was considered. This suggests full mediation whereas the indirect effect was statistically significant. Overall, the study demonstrates that the increase in academic procrastination is led by negative perfectionism due to low self-efficacy in academics.

Collectively, these studies show that academic procrastination is not just a result of poor time management. It is a behaviour that is complex which is influenced by personal, emotional and cognitive factors. Maladaptive perfectionism and self-criticism increase the tendency to postpone tasks related academics by students. Low self-efficacy, weak self-regulation, and limited resilience also make procrastination even worse. However, having better self-regulated learning higher resilience and an increase in self-efficacy in academics can help lessen these negative effects. The table below summarizes the findings clearly by outlining the key factors that affects procrastination as mentioned by the studies reviewed in this section.

Table 1: Summary of Personal, Emotional, and Cognitive Factors Influencing Procrastination Identified in Prior Studies

Factor Category	Specific Factors	(Huang et al., 2023)	(García-Ros et al., 2023)	Jing et al. (2025)
Personal Factors	Low self-regulation		✓	
	Resilience	✓		
	Self-criticism			✓
Emotional Factors	Anxiety		✓	
	Fear of failure			✓
	Low self-efficacy	✓	✓	
Cognitive Factors	Task aversiveness		✓	
	Perfectionism	✓		✓
	Metacognitive strategies		✓	

As shown in Table 1 above, procrastination is affected personal factors such as self-regulation and resilience, emotional factors such as anxiety, fear of failure, and low self-efficacy, and cognitive factors like perfectionism, and ineffective metacognitive strategies. These factors are often mentioned across multiple studies. This indicates that effective systems should be implemented to manage procrastination among students.

2.3 Machine Learning in Academic Procrastination

2.3.1 Introduction

Machine learning (ML) is a subgroup of artificial intelligence (AI) that enables computer systems to learn patterns from data and make predictions or decisions without needing explicit programming (Kufel et al., 2023). ML focuses on automating cognitive processes like decision-making and problem-solving. It allows machines to learn from experience and improve over time.

ML comprises various techniques such as supervised learning, where the models are trained using labelled data so that it can predict outcomes or categorize new observation. Another technique is unsupervised learning where it identifies the patterns or grouping in unlabelled data. Besides these two techniques, reinforcement learning allows models to learn the best course of action through trial-and-error interactions with an environment. These methods have

been used across fields like engineering, finance, medicine and many more. This is where they demonstrate their capacity to identify trends, support decision-making and forecast results from complex datasets.

ML has great potential to understand and influence human behaviour. It has been utilized extensively in healthcare to diagnose diseases, predict patient outcomes and optimize treatment plans. In education, ML can analyse students' learning habits, identify behaviours such as procrastination, and offer insights to help prompt interventions in educational settings. In a study by Baylin and Lomibao (2024), K-means clustering and filter-based feature selection were used to uncover patterns in students' procrastination tendencies and then categorized them into four different behavioural categories. The results showed that ML techniques can effectively identify procrastination levels and highlight key factors. This includes perfectionism, time management, and anxiety for maths based on this study. This emphasizes that ML has the potential to recognize procrastination-related behaviours and produce useful insights for educators.

Overall, machine learning (ML) offers a strong framework to extract meaningful information from large and complex datasets. At the same time, it also provides chances to understand, predict, and influence behaviours in various context.

2.3.2 Statistical and Structural Modelling Approaches

Research on academic procrastination have utilized numerous mathematical and statistical frameworks to measure the behaviour of students. These include models like multi-dimensional Hawkes process and Structural Equation Modeling (SEM). The main advantage of using SEM is its effectiveness in identifying the psychological factors such as anxiety or low self-regulation. These factors often correlate with task delay. Meanwhile, Hawkes process effectively models the timing and self-exciting nature of procrastination events over time. These methods provide a strong basis to understand how procrastination triggers work in a controlled environment.

The study by Jamil et al. (2021) investigated the structural relationship between procrastination and several psychological factors that are relevant to college students. This was done using a cross-sectional design with questionnaires that are self-report and using structural equation modelling (SEM) to test a causal model of procrastination. The sample included 390 students

who reported on their procrastination, difficulties in emotional regulation, depression, anxiety, perfectionism, worry, rumination and experiential avoidance. The results of SEM showed that perfectionism was affected by emotional dysregulation. This factor was not only associated with anxiety and depression but also worry, rumination, and experiential avoidance. All these factors contributed to procrastination. The results showed that worry is the strong predictor since the model fit indicates a good fit with RMSEA of 0.038 and CFI value of 0.985. The study concluded that factors such as perfectionism, negative emotions, emotion dysregulation and cognitive avoidance processes are causally related to procrastination. This suggests that addressing these factors can prove to be effective in reducing procrastination and provide a starting point for future research adopting a transdiagnostic framework.

Another study by Yao et al. (2020) used a multi-dimensional Hawkes process to investigate student procrastination in Massive Open Online Courses (MOOCs). This process models the timing of interactions between students and different learning materials such as assignments, video lectures, and discussion forums. This approach captures external factors like assignment deadlines but also internal responses where the activity sequences are triggered by previous interaction. This allows for a better understanding of patterns of procrastination. Yao et al. (2020) identified two distinct behavioural clusters which are a delay group and a non-delay group. Delay group is characterized by high excitation and bursty activity with long pauses and the non-delay group is marked by lower excitation and evenly distributed activities. The findings showed that the Hawkes model parameters represent the delays more accurately and correlate more strongly with student grades compared to traditional measures. This shows that the timing and interdependence of activities significantly impact the performance of academic. The study gathers that multidimensional Hawkes processes provide a strong framework for modelling procrastination in online learning platforms. Furthermore, this approach offers valuable insights for predicting procrastination and suggesting potential solutions.

Noviana Khoirunnisa et al. (2024) carried out a quantitative study where the authors examined the relationships between future time perspective, academic self-efficacy, academic self-control, and academic procrastination. The authors used self-report instruments and structural equation modelling (SEM) on a sample of 517 university students to test mediation effects. The study found that both future time perspective and self-efficacy on academics positively associate with academic self-control. Whereas self-control on academics significantly mediated the effects of the two predictors on academic procrastination. This indicates that students who perceive their future goals more clearly and have stronger beliefs in their

capabilities in academics tend to have greater self-control. This reduces procrastination behaviours. The SEM model showed that it has acceptable fit indices where the values of NFI, CFI, NNFI are all more than 0.90. As a result, it supports the proposed relationships among the variables. The study concluded that one of the effective interventions to target is to enhance self-control in academics as it can also help reduce procrastination by leveraging students' self-efficacy and future time orientation.

In summary, these studies provide a comprehensive overview of how SEM and multi-dimensional Hawkes process works to identify psychological drivers and temporal patterns of academic procrastination. However, these models often lack the computational flexibility and real-time scalability that is required for an application that is dynamic and gamified. Machine learning (ML) surpasses these traditional approaches by providing prediction that is accurate by using its ability to process high-dimensional, non-linear data. Consequently, using a ML-based approach is much better as it allows for a more personalized and adaptive intervention that can respond to a student's behaviour as it happens.

2.3.3 Machine Learning for Behavioural Analysis and Change

Machine learning has increasingly been used to recognize, evaluate, and predict student behaviour especially in educational settings. ML algorithms can find trends that are hidden in student engagement, learning habits and tendencies like procrastination by leveraging large and complicated datasets. These insights make it possible to create personalized interventions to support change in behaviour. These can help educators to understand how students engage with classroom settings. Machine learning is especially great for modelling complex human behaviours as it can handle versatile and nonlinear data, unlike traditional statistical methods. Lately, patterns in academic performance, engagement and motivational behaviours have been successfully identified using a variety of methods from supervised learning such as Random Forest, Support Vector Machines (SVM), and Naive Bayes to unsupervised learning such as clustering.

A study by Mohd Talib et al. (2023) predicted student success comprehensively by integrating both academic and non-academic factors such as learning program outcomes, self-efficacy, learning style and demographic information. A combination of supervised and unsupervised machine learning techniques was used to classify students into groups of low-, average-, and high- performance. Support Vector Machines (SVM) was used for classification after the K-

Means Clustering was so that it will be possible to identify the behavioural patterns of students over the course of a semester. This revealed that changes in self-efficacy and learning style had a significant impact on overall performance, especially during the final weeks. The predictive model was able to identify subtle variations in student behaviour by clustering students according to multiple factors instead of solely relying on final grades which led to increase in classification accuracy. This study proved that machine learning can identify trends in student behaviours which provides educators with useful information to implement early interventions and support techniques and strategies into practice that improve academic results and the overall development of their students.

Similarly to the previous study, a research by Shen and Yuan (2021) explored how data collected from a digital campus can be leveraged to assess and control college student behaviour using machine learning. Their research emphasized how digital solutions can collect comprehensive data about students' daily activities, consumption, and learning. The researchers hoped to gain a thorough understanding of student behaviour that goes beyond conventional management techniques. These techniques often rely on administrators' personal experience and fail to provide tailored guidance. Adaptive K-means clustering algorithm was utilized to accomplish this. This enhances the standard K-means technique by automatically adjusting the number of clusters and using the elbow rule to identify outliers. This method allowed for better classification of student behaviours in terms of consumption, lifestyle, and learning patterns more accurately and effectively. Findings from this study revealed four learning behaviour groups, three lifestyle patterns and five types of consumption behaviours. These insights can help school administrators to improve overall student management, optimize learning methodologies, and provide focused support. The research highlights the potential of machine learning to predict academic success and guide proactive, tailored interventions that promote better study and lifestyle practices.

A study by Yuan et al. (2024) investigated the integration of behavioural analysis with machine learning to evaluate online learning performance more effectively. Their research emphasized traditional predictive models that overlook the variation of students' behaviours. These variations are crucial for understanding engagement and identifying potential areas where support is needed. The researchers were able to classify students into different behavioural patterns such as driven learners, and students with poor autonomy. This classification was done by using clustering algorithms on online learning data. Random forest models and LightGBM were used to forecast learning outcomes. Crucially, the study also analysed how various

characteristics of behaviour contributed to these predictions. This enabled the teachers to identify which behaviours had the most influence on the achievement of the student. This approach accentuates how machine learning may be used not just to predict results but also to identify significant trends in student behaviour. These provide useful information to the educators to customize intervention and successfully assist the learning of the students.

In summary, the studies reviewed mentions that machine learning approaches to predict behaviour in educational settings mainly use supervised, unsupervised, and ensemble learning techniques. Supervised learning algorithms are commonly used to predict academic outcomes and identify at-risk students by analysing labelled behavioural and performance data. Supervised learning algorithms includes SVM, Random Forest, Naïve Bayes, and Gradient Boosting models such as LightBGM. Furthermore, unsupervised learning methods are often employed to reveal behavioural patterns that are hidden and divide students into different profiles of engagement or learning. Hybrid methods are also recognized in several studies, where they combine clustering with classification. It resulted in improved prediction accuracy by capturing behavioural diversity before applying predictive models. Together, these algorithms are not only able to predict the outcome but also provide a foundation for personalized and timely interventions in educational systems.

Table 2 below summarizes the key studies that was reviewed in this section where supervised, unsupervised and hybrid approaches were studied for behavioural analysis and prediction in educational settings.

Table 2: Comparison of Machine Learning Techniques Used for Behavioural Analysis and Behavioural Change in Education

Author (Year)	Educational Context	ML Approach	Algorithms Used	Behavioural Factors Analysed	Key Findings
Mohd Talib et al. (2023)	Higher education	Hybrid (Supervised + Unsupervised)	K-Means, SVM	Academic performance, self-efficacy, learning style, demographics	Clustering students before classification improved accuracy, changes seen in self-efficacy and learning style significantly influenced performance
Shen and Yuan (2021)	Digital campus behavioural management	Unsupervised Learning	Adaptive K-Means Clustering	Learning behaviour, lifestyle patterns, consumption behaviour	Identified distinct behavioural clusters, enabled tailored student management beyond traditional administrative approaches
Yuan et al. (2024)	Online learning environments	Unsupervised Learning and Ensemble Supervised Techniques	Clustering, Random Forest, LightBGM	Engagement patterns, autonomy, behavioural variation	Behavioural profiling improved outcome prediction and revealed key behavioural contributors to learning success

2.3.4 Predictive Modelling of Procrastination

The previous section 2.3.2 demonstrated how machine learning can be used to identify significant patterns in student behaviour such as student engagement levels, study habits and activity trends. Subsequently, the next step is to translate these insights into actionable predictions. Predictive modelling provides a means of identifying at-risk students before problems become worse. This is because student procrastination is often influenced by minute changes in motivation, learning behaviour, and engagement in tasks. Supervised machine learning models can predict an individual's procrastination risk more accurately by building on the behavioural patterns that were identified by clustering and other techniques that are

analytical. Hence, this section 2.3.3 focuses on research that specifically uses predictive algorithms to predict the risk of procrastination.

The study by Öztaş and Akçapınar (2025) is extremely pertinent as it investigates the possibility of accurately predicting students' academic procrastination tendencies based on their online learning behaviours. The researchers worked with 51 students from university over the course of 14 weeks to gather two types of data. The first type is a self-report questionnaire based on nine identifiable assignment-submission behaviour patterns. The second is more than 70,000 behavioural logs from Moodle activities including SCORM modules, forums, quizzes, and assignments. Features were extracted pertaining to the frequency and duration of students' interactions with course materials. Logistic Regression produced the best results with 90% accuracy in differentiating between students procrastinating and those who don't among the seven machine learning algorithms that were tested using 10-fold cross-validation. The most significant predictors were early-week engagement features indicating that students who interact regularly and early with the course materials are significantly less inclined to procrastinate. Overall, the study reveals that behavioural data can accurately predict procrastination.

Another study by Sartana et al. (2025) predicted procrastination behavior and academic performance using Moodle log data of 377 students. The study extracted features related to how frequently and regularly students participated in course activities. Indicators among the features were early engagement, delayed task initiation, and total interaction intensity. Cluster groups of three were identified by applying K-means clustering using the elbow method which are labelled as (1) low procrastination / high performance, (2) high procrastination / poor performance, and (3) medium procrastination / average performance. This was found after standardizing the data using Z-score scaling and eliminating outliers. Seven machine learning regressors were used to build the predictive models and evaluated them using the 10- and 15-fold cross validation. The machine learning regressors includes Decision Tree, Random Forest (RF), Gradient Boosting (GB), Multilayer Perceptron (MP), Support Vector Machine (SVM), K-Nearest Neighbours (K-NN), and LightGBM (LGBM). The findings showed that Random Forest performed the best with an R^2 of 0.812, a mean absolute error (MAE) of 6.248, and a root of mean square error (RMSE) of 8.456. It also revealed a strong negative correlation of -0.7777 between delayed tasks and academic outcomes. The findings emphasized that when a student delayed completing their task, lower scores were achieved. This highlights the harmful effect of procrastination. Overall, the study showed that the risk of procrastination may be

correctly predicted using behavioural engagement patterns, with ensemble-based models yielding the best performance.

Abidi et al. (2020) conducted a study to investigate how machine learning can predict procrastination by analysing students' activity patterns within an online learning management system. Rather than relying on single-event variables, the researchers identified behavioural characteristics such as when students began their assignments, how often they access course materials, and how much they delay starting required activities. The characteristics reflects the students' consistency, timeliness and frequency of engaging with course works. Ensemble machine learning models such as Logistic Regression, Decision Tree, Gradient Boosting, and Forest were used to identify and predict procrastinating students. Kolmogorov-Smirnov (KS) statistics was used as a selection criteria for classes and Average Squared Error (ASE) was used for validating data to find the best model. Gradient Boosting (GB) model that was autotuned outperformed all the other models with a KS value of 91.77% and an ASE of 2.79. The default GB model remained highly effectiveness with a KS value of 90.74% and an ASE value of 3.66. These results show that GB techniques effectively identify students at risk of procrastinating based on their online learning behaviours, especially when optimized.

Since online learning has expanded significantly during the COVID-19 pandemic, the study by Memon et al. (2020) investigated how procrastination can be pinpointed and predicted in Virtual Learning Environments (VLEs). The researchers used the Open University Learning Analytics Dataset (OULAD) that contains 32593-student profiles enrolled for courses in STEM in VLE. This dataset is used to extract the key behavioural and demographic features such as number of prior attempts made by students for the same module, final results, day of the assessment submitted, total number of credits studies and the students' assessment score to build a comprehensive dataset for procrastination. At first, clustering was intended to be used to find patterns in student behaviour. However, visualizations revealed that unsupervised approaches produced linearly separable groups and were unable to capture the complex, non-linear correlations present in tendencies of procrastination. Annotation by experts produced labels that more accurately reflected actual procrastinating tendencies considering a combination of students' involvement, performance, and submission habits. Two supervised classifiers, which are Logistic Regression and Artificial Neural Network - Multi-Layer Perceptron (ANN-MLP), were trained and optimized using the expert-annotated dataset. The ANN-MLP using Adam optimizer with a batch of 100 obtained 83.5% training accuracy and 82.9% testing accuracy whereas Logistic Regression with a similar performance obtained 82%

for training accuracy and 81.9% for testing accuracy. These results proved that the most effective method to predict procrastination is expert-guided annotation in conjunction with a well-tuned supervised model.

Table 3 below summarizes recent studies mentioned in this subsection on predictive modelling for procrastination. It highlights the features used, machine learning models applied, evaluation metrics, and the approaches that performs the best.

Table 3: Summary of Machine Learning Models and Performance Metrics Used in Prior Studies on Predictive Modelling of Academic Procrastination

Authors (Year)	Features Focused	Models Used	Performance	Best Results
Öztaş & Akçapınar (2025)	Digital traces from LMS such as course-activity logs, assignment submission patterns, timing, and frequency of access	Support Vector Machines, Decision Tree, Logistic Regression, Naive Bayes, Artificial Neural Networks, Random Forest, and K-Nearest Neighbour	Comparing Models: classification accuracy, F-score, precision, and recall, Test Model Performance: 10-fold cross validation method	Logistic Regression predicted with best accuracy of 0.90.
Abidi et al. (2020)	Interaction data from an ITS (e-learning platform) by capturing log data usage, time spent, interaction features	Logistic Regression, Tree Ensembles (Decision Trees, Gradient Boosting, Forest Model)	Kolmogorov-Smirnov (KS) statistic, Average Squared Error (ASE)	Gradient Boosting is the best model with KS and ASE value of 91.77% and 2.79% respectively.
Satrana et al. (2025)	Moodle activity logs (features were derived from timestamps of task access, submission, and engagement duration)	Decision Tree regressor (DT), Random Forest Regressor (RF), Gradient Boosting Regressor (GB), Multilayer Perception Regressor (MP), Support Vector Regressor (SV, K-Nearest Neighbours	R ² , MAE, RMSE	Random Forest (RF) achieved the highest accuracy, with an R ² score of 0.812, MAE of 6.248, and RMSE of 8.456

		Regressor (K-NN). LightGBM Regressor (LGBM)		
Memon et al. (2020)	Features focused was student_Id, Highest_Education,studied_credits, Number_of_previous_attempts ,final_result, disability, date_submitted, and score	Logistic Regression (LR)and Artificial Neural Networks (ANN)	Accuracy on training and testing	ANN training accuracy has the best model with the value of 83.3%

2.3.5 Limitations and Challenges

One of the limitations seen in the study from the previous section is that since the sample size is small with only 51 students, it restricts the extent to which the results can be applied to larger or more diverse student populations (Öztaş & Akçapınar, 2025). Even though it is common and typical for a single class to have a small sample size and is widespread in educational research. Another limitation by this study is the data was gathered during the COVID-19 pandemic. This was when students faced numerous troubles adapting to new or distant learning environments. It is challenging to generalize these findings to normal, post-pandemic conditions. This is because factors such as limited access to educational materials, increased stress, and declining motivation may have influenced their procrastinating habits. Overall, future studies should benefit from using larger and more diverse samples to improve the generalizability and the relevance of the results across various educational settings.

There were several limitations seen in the study by Sartana et al. (2025) although it presents a structured and data-driven approach. Firstly, the dataset comes from a single private university in Indonesia, hence it is limited in its applicability to other institutions, learning cultures, or structure of courses. Since the study is relying on Moodle activity logs, only the observable online behaviours were recorded. Thus, unable to take into consideration factors including psychology, motivation and other external factors that affect procrastination. Assuming that all students regularly engage with Moodle is another limitation. This is because they might be working offline, reviewing content elsewhere, or even complete their tasks without regularly using the platform, which would reduce the accuracy of the behavioural indicators. Furthermore, using heavily engineered time-based features runs the risk of oversimplifying

procrastination leading to presenting procrastination as a series of timestamps rather than a complicated behaviour influenced by environment. Since there was an absence of external validation or testing on other datasets, overfitting is a risk even though the prediction model performs well. Finally, since how the prediction can be incorporated into educational interventions were not investigated, there is a gap between algorithm output and real-world application.

The study by Abidi et al. (2020) acknowledges several limitations could restrict the applicability of its conclusion. It still depended on a limited set of behavioural variables although many other relevant indicators such as delaying of tasks, levels of interaction, or time spent engaging with learning materials could have been included to build a more comprehensive model. Although autotuned algorithms introduce constraints since choosing hyperparameters automatically does not ensure the finding of the truly optimal model, even though it offers a starting point for further refinement. The study could have improved prediction accuracy if mental and health problems of the students were taken into consideration. Future work would probably increase the model's robustness and provide a more comprehensive knowledge of procrastination patterns by including a wider range of behavioural, contextual, and wellbeing-related characteristics.

One of the other limitations seen by the study by Memon et al. (2020) relied solely on a single secondary dataset (OULAD). This dataset may not fully represent student behaviour across various institutions, cultures, or online learning platforms. Detailed guidelines by the expert-driven label are still subjective and based on assumptions that might not hold true for all students. Hence, it might introduce bias into the classification of procrastination. The clustering method and visual comparisons also highlight that many of the patterns from the data are unclear or overlapping. Thus, making it challenging to tell procrastinators from non-procrastinators with confidence.

Table 4 below presents the focus of the research, methodologies, and limitation that were identified when discussing the studies in this section.

Table 4: Summary of Related Studies on Predictive Modelling of Academic Procrastination and Their Identified Limitations

Author (Year)	Research Focus	Key Findings & Methodology	Identified Limitations
Öztaş and Akçapınar (2025)	Prediction of academic procrastination tendencies	Used behavioural trajectories in online learning traces to classify students	Small sample size (n = 51) Data collected during COVID-19 may lack external validity due to abnormal learning patterns
Sartana et al. (2025)	LMS-based prediction using machine learning	Random Forest and L-Means clustering applied to Moodle logs	Single institution context Relies only on observable logs Lacks translation of algorithmic results into active interventions
Abidi et al. (2020)	Behavioural data-driven performance prediction	Applied autotuned algorithms to behavioural variables and procrastination levels	Limited variable set Excluded task delay patterns and psychological factors that influence behaviour
Memon et al. (2020)	Learning analytics-based behaviour analysis	Utilized secondary datasets to analyse student interaction and procrastination.	Relies solely on a single secondary dataset (OULAD) which restricts generalizability across different systems of educations

2.4 Gamification and Its Role in Behaviour Change

2.4.1 Concept of Gamification

Gamification has become a popular strategy to improve results of learning, enhance motivation and increase participation in various fields, especially in education. Gamification, which derives from principles of game design, is the process of introducing game-like elements such as points, badges, levels, and challenges. These elements are incorporated into a non-gaming environment to promote engagement and persistence. Gamification as a method has become more common to address issues like low motivation, procrastination, and inconsistent engagement in the learning environment. This is due to the growth of digital technology and online platforms.

Hebebcı and Alan (2021) defines gamification as the application of game design features in a non-gaming environment. This was first proposed by Nick Pelling in 2002. In 2008, the idea gained more attention, and it became even more popular after the 2010 Design, Innovate,

Communicate, Entertain conference. Applications like digital badges, and points systems sparked interest in gamification marketing, trade and advertising. Gamification has gradually expanded into educational environments. For example, platforms like ClassDojo and Khan Academy use virtual characters, badges, and progress tracking to improve student involvement and track their performance. This demonstrates how gamification uses game mechanics to improve engagement and motivation in both business and educational contexts.

However, Staller and Koerner (2021) argue that gamification lacks a widely agreed definition and they emphasize the differences and similarities between play and games. They note that educational gamification involves both experiential factors like joy, engagement, and motivation as well as structural elements like rules and mechanics. The authors encouraged educators to take a reflective approach that considers context-specific factors, the visibility of gamified aspects, and students' acceptance in their "non-definition" of gamification. From this perspective, gamification is a flexible and adaptable tool that can be utilized to enhance engagement and persistence in learning. Its effectiveness not only depends on game mechanics but also on how the students view and interact with the gamified environment.

Gamification in eLearning systems should follow a clear framework based on instructional design principles (Yamani, 2021). The study identifies key components of gamification such as points, rewards, feedback loops, and progression mechanics. It discusses how their organized use might improve learner motivation and participation. Gamification can create a sense of achievement, encourage active participation, and provide ongoing rewards by integrating these components into course design. The framework also emphasizes the importance of user-centered, game-centered, and technology-centered methods to ensure that gamification aligns with the learners' aims and goals. This structured approach can improve the information retention, promote skill development, and create an environment that minimizes negative habits like procrastination.

In summary, gamification combines game-based tactics with behavioural and educational goals to increase motivation, engagement and learning effectiveness. It also offers a flexible framework to influence the learners' behaviour by using psychological concepts, structured game mechanics, and historical roots. According to the research, gamification has strong potential to address academic issues like procrastination. Gamified interventions encourage timely engagement and consistent study habits by offering rewards, progress tracking, and enabling interactive learning.

2.4.2 Gamification for Procrastination

Recently, gamification has gained more attention because it has become a potential strategy to control and overcome procrastination. Gamification seeks to increase initiation of tasks and sustained effort by leveraging features such as points, rewards, and many more game elements. This method is appealing as it may have the ability to transform effortful or difficult tasks into experiences that are more organized and fulfilling. Researchers have explored how different types of game elements can strengthen commitment to tasks that are scheduled, promote regular productivity behaviour, and assist self-regulation as digital platforms continue to develop. Therefore, this section will look into some of the studies that examines how gamification has been created and used to lower procrastination in academic settings.

Pereira et al. (2025) carried out a thorough literature analysis to investigate how gamification has been applied to procrastination and the true efficacy of these tactics. Their research synthesizes results from other numerous studies where they demonstrate that although gamification is often used to increase motivation, elements like points, badges, leaderboards, and progress bars make the results inconsistent. These game mechanics might help make activities more engaging and manageable particularly in academic settings. However, their efficacy mostly depends on how the elements are created and whether they activate intrinsic motivation instead of only depending on rewards from external sources. This study also emphasizes that many interventions have strong focus on extrinsic motivators, which is great for improvements for a short-term but not in the long-term as it may lose its effectiveness. Crucially, the authors stress that to create gamification systems that result in significant and long-lasting behavioral change, personalization, thoughtful design, directly addressing the underlying cause of procrastination are essential. Underlying cause of procrastination refers to the low motivation, poor emotional regulation and time management.

Another study by Amit et al. (2021) presents a useful, gamified framework intended to decrease procrastination by incorporating several motivational techniques into a mobile app called ProScore. The system creates a structured yet engaging environment that encourages starting task and maintained focus by combining a modified Pomodoro timer with game elements like points, leaderboards, and live activity streams. By dividing work into manageable intervals, the Pomodoro approach creates a sense of urgency, while the gamification layer reinforces productive behaviour by giving points to the users for finishing sessions and penalize them

when they stop doing the activities in the middle. The leaderboard increases peer influences and uses social comparison to combat procrastination by motivating the users to improve their ranks and match the productivity levels of their peers. The system also includes personalized task-duration estimates, which helps users determine realistic time requirements based on past data. This aims to reduce planning uncertainty, which is a common cause of procrastination. The framework from this study shows that combining time-management strategies with gamification, incentives, and social accountability can lead to an engaging, user-friendly intervention that helps individuals overcome procrastination and maintain consistent productivity.

A recent research by Sun (2024) studied the app called Forest: Focus for Productivity to explore how gamification affects academic procrastination. The study involved surveys and interviews with five active users of the app. It investigates how gamification elements including rewards, punishments, personalization and collaboration affect the behaviour of procrastinating. The study found that the gamification significantly improves the user experience and supports the app's primary intervention strategies rather than directly enhancing motivation. Although the cooperation and punishment helped the users stay focused and restrain impulsivity, rewards and personalization improved user satisfaction and the sense of achievement. Participants noted that the core intervention tactics play a key role in the app's effectiveness. This is because gamification elevates the significance and engagement of the experience.

In summary, gamification effectively boosts user engagement and offer meaningful support in managing academic procrastination. This is feasible by offering rewards, collaboration, customization, and other interactive elements. Game elements mainly enhance the effectiveness of intervention measures rather than directly addressing procrastination. However, its impact varies based on motivations and characteristics of each individual user. Despite some limitations, the studies mentioned in this section showed the potential of gamified application to create more engaging and dedicated learning experiences. This suggests that careful integration of gamification can be a useful tool for addressing procrastination.

The table below reviews the gamification approaches in relation to procrastination and productivity. It details the specific game elements implemented and their effects on engagement and motivation.

Table 5: Summary of Prior Studies on Gamification Elements Applied to Procrastination and Productivity Management

Author (Year)	Context	Gamification Elements Used	Key Findings
Pereira et al. (2025)	Systematic review of gamification in procrastination / productivity contexts	Common elements aggregated across many studies: Points, Badges, Leaderboards, Progress Bars, Challenges, Avatars, Narrative, Friend Interaction / Social, Group Competition, Mini-games, Levels/Phases, Visual representations, Resource Management, Chatbots	Points/Badges/Leaderboards/Progress Bars (“BPL + Progress”) are most used. Social, narrative, and collaborative elements much less frequent. suggests over-reliance on extrinsic motivators.
Amit et al. (2021)	Online learning environment (assessing which gamification components improve engagement)	Standard gamification components: likely including progress tracking, points/badges, maybe feedback (details as typical gamification toolkit.)	Showed that certain gamification components correlated with higher engagement in online learning. Supporting that well-chosen mechanics can enhance user interaction.
Sun (2024)	Case study of productivity app targeting academic procrastination	“Forest”-style gamification: reward/penalty for focus sessions (gamified timer), virtual “planting” metaphor, visual forest, progress tracking, collaboration/ co-planting, unlockable content / customization, achievement / reward, social/ group accountability	Gamification helps with focus and engagement, but effectiveness depends on user motivation. Gamification alone not a guaranteed fix.

2.4.3 Limitations of the Gamified Approaches

Even though Pereira et al. (2025) provide valuable information about the connection between gamification and procrastination, the study also notes several limitations that should be acknowledged. Since the study is a systematic literature review, the findings solely depend on the quality, consistency, and thoroughness of the included studies. Many of these studies may vary in design, sample size, context and measurement of procrastination. This makes it challenging to draw firm conclusions about which gamification techniques are successful or under what condition they work the best. The review also points out that a significant number

of existing studies focus on short-term intervention. This means that long-term behavioural change and the sustainability of gamification effects are still unknown.

Next, a limitation of the ProScore study is that the framework may limit its effectiveness since individual and environmental factors are not fully accounted for (Amit et al., 2021). Although the Pomodoro technique, peer pressure, gamification, and incentives are all integrated in the application, the responses by the users may vary greatly based on their levels of motivation, personality attributes, and past time management experience. Furthermore, the reliance on user-defined task durations and arithmetic mean calculation to estimate time may limit the accuracy of task planning, especially for complicated or highlight variable tasks. Since historical data is required to enhance time estimates, the framework also faces issues for new users, which could reduce initial usefulness.

As for the study Sun (2024), there are also a few limitations that need to be acknowledged. Firstly, the study only consists of a very small sample size of five participants making it challenging to draw generalization conclusions about the broader user population. Purposive and convenience sampling reduces representativeness because participants with very high engagement or extreme usage habits may skew the results. Additionally, the study mostly relied on self-reported data from surveys and interviews, which can introduce biases like individuals giving socially acceptable answers or errors in memory recall. As the study only focuses on the Forest app, the conclusion might not accurately reflect how gamification works in other productivity apps where there are different user bases and game designs. Finally, even though the study effectively investigated user experiences and opinions, the sequencing of survey and interview analyses may have been enhanced to find and explore interesting patterns in a more systematic manner among the participants. These limitations imply that the results provide preliminary insights rather than offering conclusive proof of the efficacy of gamification in lowering academic procrastination.

Table 6 below summarizes the key limitations that was identified by the studies reviewed in this section on examining gamified approaches to address academic procrastination.

Table 6: Key Limitations Identified in Prior Studies on Gamification-Based Procrastination and Productivity Interventions

Author (Year)	Context	Key Limitations Identified
Pereira et al. (2025)	Systematic literature review on gamification and academic procrastination	Studies included varied widely in research design, sample size, educational context, and methods used to measure procrastination. Most studies focus on short-term outcomes on interventions.
Amit et al. (2021)	Gamified productivity and time management application	Experience such as motivation, personality traits, and prior time management not considered as everyone are different. Task duration estimates rely on user input and simple averaging methods, thus reducing accuracy for complex or variable tasks. The study uses historical data which limits effectiveness for new users.
Sun (2024)	Qualitative study of the Forest productivity app	Uses a small sample size of 5 participants. Purposive and convenience sampling limits representativeness. Data relies heavily on self-reported surveys and interviews that can lead to introducing bias. The focus on a single app restricts generalisability to other gamified systems.

2.5 AI-Based Hierarchical Task Decomposition and Planning

The decomposition of complex tasks into manageable subtasks is one of the long-standing problems in artificial intelligence (AI). Notably, this problem is within planning and decision-support systems. Decomposing tasks hierarchically enables the system to transform high-level, abstract objectives into sequences of executable actions that are structured. This enhances interpretability and feasibility in complex environments. Recent research has advanced in this area by combining traditional hierarchical planning techniques with optimization methods and adaptive AI reasoning. This highlights how the relevance of such approaches for application that need intelligent task breakdowns and adaptive support.

A study by Elkawkagy et al. (2025) proposed using Hierarchical Task Network (HTN) and combining it with Ant Colony Optimization (ACO) to create an integrated framework during the refinement phase. Candidate task decompositions are evaluated and refined using swarm-intelligence principles using the ACO-HTN approach. This allows the planners to navigate large solution spaces more effectively. Both quality of the solutions and computational efficiency provide improvements in their empirical results compared to HTN planners. This study emphasizes the value of rule-based hierarchical decomposition enhanced with adaptive

optimization mechanisms, especially for complex tasks. Naive decomposition may lead to inefficient or suboptimal plans.

The next study by Kienle et al. (2025) introduces LODGE which is a framework that jointly performs hierarchical task planning and domain model learning. This framework is supported by grounded execution and simulation. LODGE learns hierarchical abstractions that connect low-level actions to higher-level tasks instead of relying solely on status, hand-crafted planning rules. Meanwhile, LODGE also validates their feasibility through execution feedback. If there are long horizon planning problems, then this hierarchical structure is very effective whereas flat planning approaches often fail. This study demonstrates that combining structured task decomposition with consistency checks across planning levels significantly improves planning success rates. Thus, reinforcing the importance of hierarchical representations for complex task management systems.

ADAPT framework which is an as-needed decomposition framework was proposed by Prasad et al. (2024) which dynamically breaks down tasks based on execution difficulty. ADAPT recursively decomposes subtasks only when execution fails thus allowing the system to adjust decomposition depth according to the complexity of the task and the capability of the system. This approach differs from fixed hierarchical planners, which decompose tasks upfront. This adaptive decomposition strategy has proven to outperform traditional planning and executing tasks across multiple benchmark environments. The findings from this study suggest that effective task breakdown does not need exhaustive upfront planning. Instead, it needs context-aware decomposition that responds to execution constraints.

In summary, the studies mentioned demonstrate that hierarchical task decomposition is a strong method for managing complex tasks in intelligent systems. Approaches like optimized rule-based planning, learned hierarchical domain models or adaptive as-needed decomposition strategies, highlight the importance of structured task breakdown in supporting execution, decision-making, and user guidance. These insights gained from these studies provide a strong theoretical foundation to integrate AI-driven task decomposition mechanisms into student-focused app. This is where the large academic tasks must be broken down into smaller, actionable subtasks to reduce cognitive overload and procrastination risk.

The table 7 below presents representative AI-based hierarchical task decomposition approaches based on the studies explored in this section and their relevance to managing tasks that are complex.

Table 7: Summary of Task Decomposition Approaches and Their Key Contributions

Author (Year)	Proposed Approach	Key Contributions
Elkawagy et al. (2025)	Integration of Hierarchical Task Network planning with Ant Colony Optimization	The framework refines task decomposition using swarm intelligence principles as it improves solution quality and computational efficiency compared to traditional HTN planners.
Kienle et al. (2025)	LODGE framework combining hierarchical task planning and domain model learning	The framework learns hierarchical abstractions from execution and simulation rather than relying solely on manually defined rules as it improves planning success for long horizon tasks.
Prasad et al. (2024)	ADAPT framework with as-needed task decomposition	Tasks are decomposed dynamically only when execution fails. This allows decomposition depth to adjust based on the complexity of the task and capability of the system. This approach outperforms fixed decomposition strategies.

3.0 Methodology

3.1 Overview

This chapter explores the systematic approach that will be used to develop the gamified app. This project adopts a dual-pipeline methodology. This is because the machine learning pipeline and the development of the application pipeline will be executed in parallel and will be integrated into a system that is unified later. This design allows the development of the predictive model and the application to simultaneously progress.

The machine learning pipeline focuses on preparing the secondary dataset from the raw dataset. This is to engineer only relevant features, and to train multiple machine learning models. Then the performance from the models will be evaluated to identify the most suitable model to predict the procrastination risk. At the same time, the application development pipeline will concentrate on analysis of requirements, design of the app and the gamification elements, frontend and backend development, and finally the preparation of an application prototype.

The pipelines after meeting the predefined criteria, will be merged in the final phase which is integrating into the system. This is where the best trained machine learning model is deployed as an API and connected to the application backend. This will be enabled to predict the risk of procrastination all while supporting the adaptive and gamified adaptive feedback delivery.

The dual-pipeline approach was adopted to improve project development by allowing both the machine learning model and prototype for the application to progress at the same time. This method ensures that the predictive model can be tested and refined repeatedly. At the same time, the applications interface and the features for gamification are developed. Combining the pipelines at the end creates flexible system. This allows updates to either the models or the application and allows improvements to one without affecting the other.

3.2 Process Flow

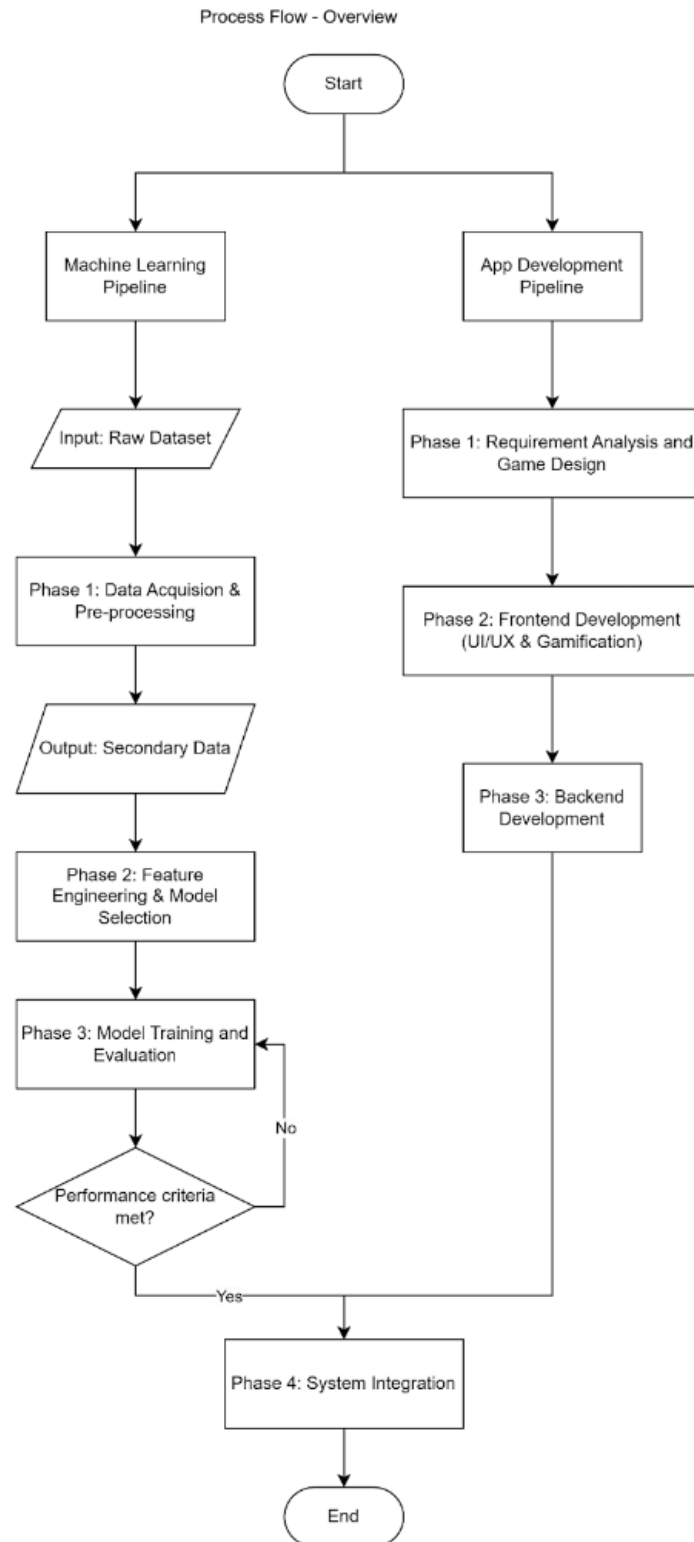


Figure 1: Process Flow Diagram

The Figure 1 above represents the overall process flow diagram of the methodology section that was proposed. The diagram illustrates the two pipelines that are parallel, which are machine learning pipeline and application development pipeline. Both the pipelines will be initiated simultaneously to support efficient and coordinated development. The first pipeline which is the machine learning pipeline will comprise of data pre-processing, feature engineering, and model training with performance criteria that is predefined. The second pipeline which is the application development pipeline will focus on requirement analysis, gamification design, frontend and backend implementation. This is to create a functional application prototype. When both the pipeline is completed, the machine learning model that was trained will be integrated into the application in the final phase which is the system integration phase. This will enable the app to predict procrastination and will provide gamified feedback delivery.

3.3 Dataset Description

The dataset that is used in this project was obtained by Rusdi et al. (2020) by collecting the academic procrastination data of undergraduate students at several universities in Makassar, South Sulawesi, Indonesia. A structured questionnaire based on the Procrastination Assessment Scale-Student (PASS) was to collect information from 586 students where 310 of the students are males and the rest of 276 students are females. Five-point Likert scale ranging from strongly disagree to strongly agree was used to allow quantitative measurement of procrastination-related behaviours and the reasons behind them.

This dataset was chosen because it captures the behaviours associated with academic procrastination and the psychological factors behind it. This makes it ideal for modelling procrastination risk. This dataset includes two main section where the first section focuses on areas of academic procrastination whereas the second section measures two factors which are Fear of Failure and Task Aversiveness. The first factor which is Fear of Failure prioritizes anxiety about meeting the expectation of one's own and other people's standards, fear of achievement, and lack of self-assurance. This section includes 18 items to determine the level of students' academic procrastination behaviour. The second factor which is Task Aversiveness highlights the unpleasantness of tasks, willingness to take risks, and lack of assertion.

Additionally, the dataset is from a peer-reviewed and publicly available data article. The article reports construct validity, factor loadings, and high internal reliability with Cronbach's alpha values over 0.90. This supports reliability and theoretical foundation of the data. The factors evaluated in this dataset are widely accepted in academic procrastination research and can be applied across various higher education settings although the data was collected in universities from Indonesia.

Therefore, this dataset helps to identify procrastination by highlighting common patterns, which in turn allows to make the system smarter and more responsive to user needs. Findings obtained from this data will help develop personalized nudges and feedback mechanisms within the gamified mobile application. Hence, allowing the system to respond effectively when tendency of procrastination is identified.

Table 8 outlines the structure of the dataset for this study which includes the demographics, target measures of procrastination and input features.

Table 8: Summary of Dataset Variables and Their Descriptions

Section	Variable Name	Items	Description
Demographics	SEX	N/A	Gender of the student (Male/Female)
	UNIVERSITY	N/A	Which university the student is from (Makassar State University, Hasanuddin University and the Department of Psychology, Makassar Satte Islamic University)
Section 1	Areas of Procrastination	A1 - A18	Measures frequency and problem severity across 6 tasks (Used as Target)
Section 2	Reasons for Procrastination	R19 - R44	Measures underlying causes such as anxiety, laziness, or rebellion (Used as Features)

3.4 Machine Learning Pipeline

3.4.1 Phase 1: Data Pre-processing

This initial phase's objective is to ensure the integrity of the raw data before performing the next step which is feature selection. This phase was done to reduce the high dimensionality of the raw survey where it has 44 items.

The raw dataset by Rusdi et al. (2020) will undergo the processes below:

1. Treating the Missing Values: The dataset will be examined for incomplete records. Rows that contain null values in crucial columns will be removed to preserve the validity of the statistics.
2. Validation of Outlier and Range: The dataset contains ordinal data as it uses the 5-point Likert Scale. So, anomalies entries will be removed so that all the values will strictly fall into the standard range of 1 to 5. This will help validate the responses.
3. Check for Consistency: Demographic label (Gender) will be standardized to make sure there is consistent encoding during the analysis.

These preprocessing steps are crucial to ensure the quality of the dataset and keeping valid statistical relationships intact. By cleaning the missing values and standardizing ordinal variables can help the model to generalize better and reduce bias caused by inputs that are inconsistent.

3.4.2 Phase 2: Feature Selection and Secondary Dataset Construction

This second phase focuses on transforming the cleaned raw dataset into a refined secondary dataset. The raw dataset which initially had 44 items was transformed into a streamlined feature set with only 11 variables. This is to eliminate the statistical noise and prevent the data leakage. Using the entire 44 variables to train the model will be unsuitable due to the redundant variables, overfitting and the risk of leakage of data. Hence, systematic feature selection that focuses on predictive psychological drivers of procrastination was applied.

The target variable was selected by using the A7 item from the original dataset. The description of the item 'A7' is "To what degree do you procrastinate this task?". The task here refers to weekly reading assignments. The variable was chosen as the single target variable for the model as it revealed that A7 measures not only the continuous but also the habit that is highly occurring. The other variables such as A1, A4, A10, A13, and A16 where it focuses more on tasks that happen occasionally such as writing term papers or studying for exams. All the other variables except for A7 will be excluded in the secondary dataset.

Next, the input variables will be identified to analyse the psychological drivers of the target behaviour. The input variables will be chosen from the items R19 till R44 where it focuses on the reasons of procrastination. The items were chosen based on the importance score using Random Forest Feature.

Table 9: Top 10 Features Selected from the Raw Dataset Based on Importance Scores

Question	Importance Score
R-28	0.047333
R-37	0.047138
R-40	0.046400
R-36	0.045759
R-38	0.045131
R-22	0.045011
R-32	0.041275
R-43	0.041013
R-42	0.040591
R-44	0.040563

Figure 2: Feature Importance Scores for the Top 10 Variables Selected from the Raw Dataset

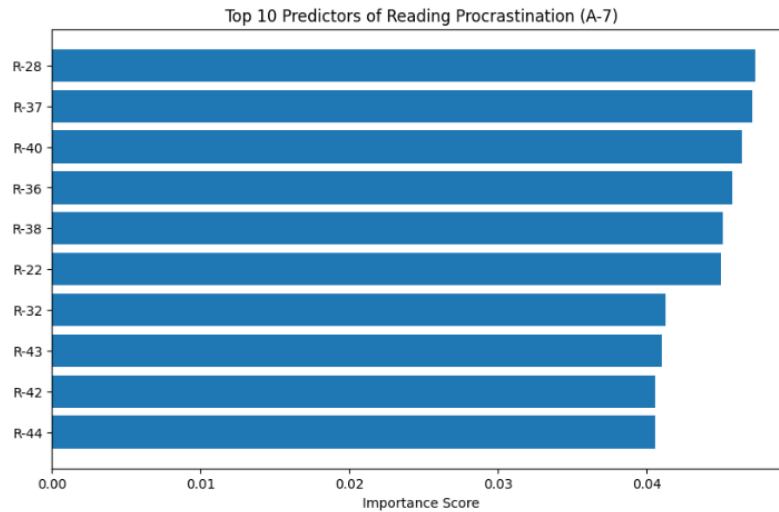


Table 9 and Figure 2 above show the top 10 variables with the highest important scores, and these were selected as the final input variables. The remaining 16 items were removed to reduce the model noise and improve the computational efficiency.

3.4.2.1 Final Structure of Secondary Dataset

The final processed dataset contains 1 target variable and 10 input variables. This dataset will be used further on for machine learning models in the next phase.

Below is the table that shows the variables chosen, its role, new name for the variables chosen and its description.

Table 10: Final Variables Selected for the Secondary Dataset with Roles and Descriptions

Variable Role	Original Variable Name	New Variable Name	Description
Target	A7	Target_Risk_Class	Binary Target
Input	R28	Overwhelm_Score	Ordinal (1 - 5)
Input	R37	Peer_Pressure_Score	Ordinal (1 - 5)
Input	R40	Fear_Success_Score	Ordinal (1 - 5)
Input	R36	Sensation_Seeking	Ordinal (1 - 5)
Input	R38	Rebellion_Score	Ordinal (1 - 5)
Input	R22	Time_Mngt_Score	Ordinal (1 - 5)
Input	R32	Social_Anxiety_Score	Ordinal (1 - 5)
Input	R43	Low_Energy_Score	Ordinal (1 - 5)
Input	R42	Perfectionism_Score	Ordinal (1 - 5)
Input	R44	Social_Distracton	Ordinal (1 - 5)

3.4.3 Phase 3: Model Training and Evaluation

With the feature selected from the secondary dataset in Phase 2, the next phase focuses on the experimental framework used to train, validate and test the predictive models. The objective of the third phase is to identify the algorithm that yields the highest reliability for real-world application.

3.4.3.1 Data Partitioning

The dataset will be partitioned using an 80/10/10 split to ensure robust model development and also unbiased performance prediction. The split consists of 80% of training, 10% of validation and 10% of testing. The training set will be used to learn the model parameters while the validation set will be employed for hyperparameter tuning and model selection, and finally the test set is restricted for final performance evaluation. This split will be adopted into my project to prevent the leakage of information and to also maintain fair comparison among the models chosen.

3.4.3.2 Model Selection

The machine learning models chosen below are based on the highest reported accuracy in prior studies from the literature section in Chapter 2. This is to adopt a context-driven model selection strategy. In the section of 2.3.3 in Chapter 2, four studies were examined on predicting academic procrastination, where each of the studies employed different selected models all according to the characteristics of the data, interpretability requirements, and intended application context.

The first model that will be adopted is Logistic Regression which is a baseline model as it was frequently used in the research where questionnaire-based psychological data are involved. This model was selected in the existing studies for its interpretability and its ability to quantify the influence of individual psychological factors. In this project, this model will serve as a transparent reference model that allows comparison between the statistical learning that is traditional and more complex approaches. Since the target variable is a binary variable, this model is very suitable to be used. Also, this model has high interpretability for psychological constructs and could also act as a benchmark for model comparison.

The next model selected is support vector machine (SVM) because of its effectiveness when handling high-dimensional, and datasets that are small-to-medium-sized, which is a characteristic of questionnaire-based behavioural data. SVMs are very suitable especially when the class boundaries are non-linear and not easily separable using simple linear models. SVM is very suitable to be used on the secondary dataset due to its effectiveness for Likert-scale psychological features, robust overfitting in medium-sized datasets and appropriate for binary classification tasks.

Decision trees are also included in this ML approach to provide a rule-based model that is also human-readable. This model mirrors decision-making logic often used in behavioral interventions. Decision trees have structures that are highly interpretable, capable of modeling nonlinear relationships, and are useful for translating predictions into gamified rules and feedback logic.

The fourth model chosen is the Random Forest which was often used in the existing studies due to its ability to capture nonlinear interactions among procrastination-related factors. The Random Forest model is well aligned with the theoretical part of the problem in the project since procrastination behaviour often arises from complex interactions between emotional, cognitive, and environmental triggers. Random Forest is one of the chosen models as it can

handle nonlinear relationships without explicit feature engineering, durable to noise and multicollinearity and is also able to provide feature importance insights relevant to behavioural interpretation.

The next model chosen is Gradient Boosting Model such as XGBoost or LightGBM as it can increasingly improve weak learners and make them suitable to capture subtle behavioural patterns. The previous studies adopted this model when predictive performance was prioritized under structured tabular data. This model has a very high predictive capability for tabular psychological datasets, effective at modelling complex decision boundaries and is also suitable to predict risk systems requiring reliable classifications.

Another model that will be used for this project is k-nearest Neighbours (k-NN) as it can be added as a similarity-based learning model. This aligns conceptually with behavioural comparison among students with similar procrastination profiles. This model is great as it is intuitive distance-based classification and is useful for behavioural similarity analysis and also provides contrast against parametric and ensemble models.

A Multilayer Perceptron (MLP) will also be included to evaluate whether this neural network-based method can capture the nonlinear latent relationships among psychological factors that traditional may not fully represent. This model is very suitable for behavioural pattern learning and it allows comparison between the traditional ML approaches and neural approaches.

3.4.3.3 Model Evaluation Metrics

The model performance will be evaluated using multiple metrics so that the assessment is ensured to be balanced. The metrics include accuracy, precision, recall, F1-score, and confusion matrix. Since the context of the application is to detect the risk of procrastination earlier, it particularly emphasizes on the recall metrics. Hence, failing to identify the at-risk student may reduce the effectiveness of the gamified intervention.

3.4.4 Application Development Pipeline

This section will focus on the pipeline on developing the application where the users can use it to manage their tasks, assignments and many more. This application will include gamification elements to make it more interactive with the users. This application pipeline emphasized the

usability, engagement, and behavioural change through the design of the app that is well-designed.

3.4.4.1 Requirement Analysis and Gamification Design

The requirement analysis phase is where the functional and non-functional expectations of the procrastination intervention system will be established. The functional requirements define what the system must do which includes registration of the user, data collection through questionnaires, procrastination risk prediction, and delivery of the personalized gamified feedback. System performance usability, scalability, data privacy, and responsiveness will all fall under the nonfunctional requirements. This is to ensure that the application can support sustained student use without performance degradation.

The key procrastination factors from the literature such as low self-regulation, task aversiveness, emotional avoidance and fear of failure, are linked to specific gamification strategies following the identification of requirements. This link will ensure a clear relationship between psychological triggers of procrastination and the motivational aspects of gamification. For instance, breaking tasks into smaller steps will help tackle task aversiveness whereas the achievements will help address low self-confidence of the users.

Specific game elements are chosen, and their use will be explained based on this connection. Points will encourage consistent engagement, badges will act as a reward for ongoing improvement, streaks promote regular task engagement, and timely prompts and nudges will encourage action when the user potentially procrastinates or is procrastinating. These gamification elements are chosen not only for engagement but also for their ability to help with self-regulation and maintaining behaviour.

3.4.4.2 Frontend Development

The frontend development will focus on creating a user-centered and intuitive interface that will support interaction that is continuous with the system. A UI/UX design approach that is minimalist will be adopted to reduce the cognitive overload, especially for users that are already experiencing procrastination-related stress. The design will include visual hierarchy, colour schemes that are consistent, and apps navigation that are clear so that it is useful to help to guide the user's attention toward the elements.

The user interaction flow follows a natural cycle which is the onboarding process, doing self-assessment, receiving feedback or nudges, and monitoring progress. Users start by registering themselves if they are a first-time user of the app and then will start completing a questionnaire and then receive a procrastination risk status along with contextual feedback. Later interactions in the app will let the users track their progress, see achievements, and respond to system prompts.

Feedback and progress visualization are crucial for keeping users engaged. Dashboards show procrastination risk status, streak continuity, and accumulated achievements in a format that is easy to read. Visual indicators will be designed to highlight the improvement trends rather than negative outcomes to promote positive behaviour change instead of feelings of guilt or failure due to procrastination.

Besides this, the frontend will also support AI-assisted task breakdowns where users can submit their academic tasks with either short or detailed descriptions, and then the task breakdown will be generated. The task breakdown will be organized hierarchically where it shows the main tasks, subtasks, and deadlines in a clear format. This structure keeps the AI-generated plans easy to understand and act on. It helps reduce mental strains and keeps the users engaged without overcomplicating the system.

3.4.4.3 Backend Development

This phase will focus on managing the system logic, handling data, and ensuring the communication between the system components. A well-structured database will be created in the backend to store the user profiles, responses to the questionnaire, prediction outputs, and gamification states such as points, streak, and badges. Data normalization and indexing will ensure efficient retrieval and updates.

The user data will be handled with privacy in mind. Private information will be separated from behavioural data, and the sensitive records will be protected by the controls in place. All interactions between the frontend and backend will be managed through secure API endpoints to maintain data integrity during transmission.

System logic will indicate how the prediction will be processed and how gamified responses are created. When new user data is inputted, the backend will manage the data preprocessing and then the prediction model will be called upon, which later will check the level of risk and

determine the appropriate feedback or gamified response. This logic that is centralized will ensure consistency across different user sessions and contexts.

Rule-based planning approach will be used for the AI-based task decomposition logic in the backend. The task descriptions that will be submitted by the user will be turned into structured execution plans. The system will examine the task scope, workload, time limits to create subtasks and set internal deadlines before the actual due date. The backend is able to support the consistency, reusability, and the scalability of the planning capabilities by keeping the task decomposition logic separate from the presentation and storage layers.

3.4.4.4 User Interface Design

Below is the user interface design of the proposed mobile application. This section illustrates the layout, navigation structure, and interaction flow between screens. The diagrams below provide a visual representation of the interaction between the user and the system.

Figure 3: Login page and new user registration interface.

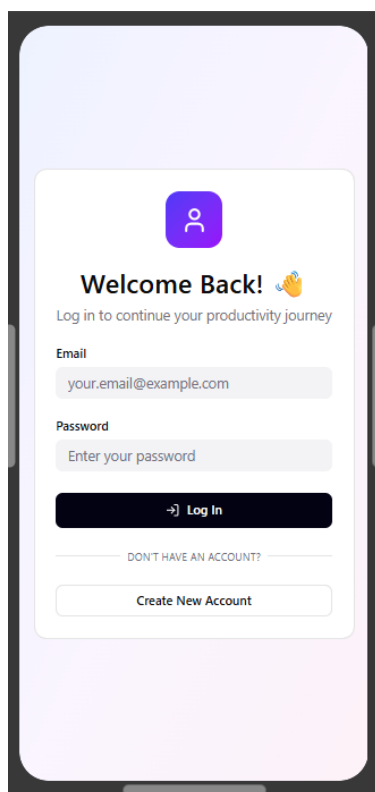


Figure 4: User registration form for creating a new account

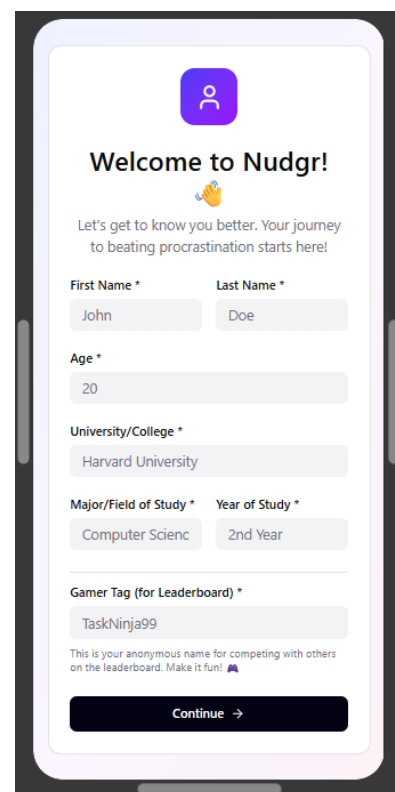


Figure 5: Procrastination and academic behaviour survey used to assess user profile and risk factors

Question 1 of 7

How would you rate your procrastination level?

Be honest! This helps us personalize your experience.

1 - Never procrastinate 5 10 - Always procrastinate

Next →

Figure 6: Main task page showing options to create tasks

Welcome to Nudgr!
Your personalized productivity journey starts now!

✓ You're on track!
Great job staying ahead of your deadlines.
Progress over perfection! ★

Active (0) Completed (0)

No active tasks
Create your first task to get started!

+ New Task

Tasks Leaderboard Account

Figure 7: Pop-up interface displayed when creating a new task from the main task page

My Tasks
0 active tasks

+ New Task

Create New Task

Add a new assignment or task. Set an internal deadline to stay ahead of schedule.

Task Title *

e.g., Write essay on climate change

Description (optional)

Add any additional details, instructions, or requirements for this task. The more details you provide, the better AI can break it down into subtasks.

Tip: Include assignment instructions or requirements for better AI task breakdown!

Course * Priority *

Select course Medium

Internal Deadline * Actual Due Date *

mm/dd/yyyy mm/dd/yyyy

Your personal goal date Official assignment due date

Create Task

Cancel

Tasks Leaderboard Account

Figure 8: Task page after a new task is created, including feedback alert for task creation

Task created!
Don't forget to set your internal deadline to stay ahead!

✓ You're on track!
Great job staying ahead of your deadlines. You're capable of amazing things! 🍌

Active (1) Completed (1)

☐ Physical - Build Model high

Build model based on Newton's Law

120 pts (2x bonus!)

Physics

Internal Deadline Actual Due Date
2026-01-05 2026-01-10

Break down into subtasks

Tasks Leaderboard Account

Figure 9: AI-powered task decomposition view showing subtasks

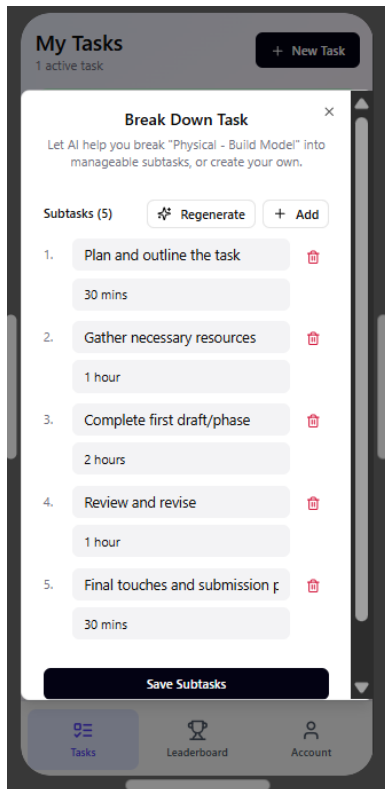


Figure 10: Points and rewards display indicating user progress after completing a task

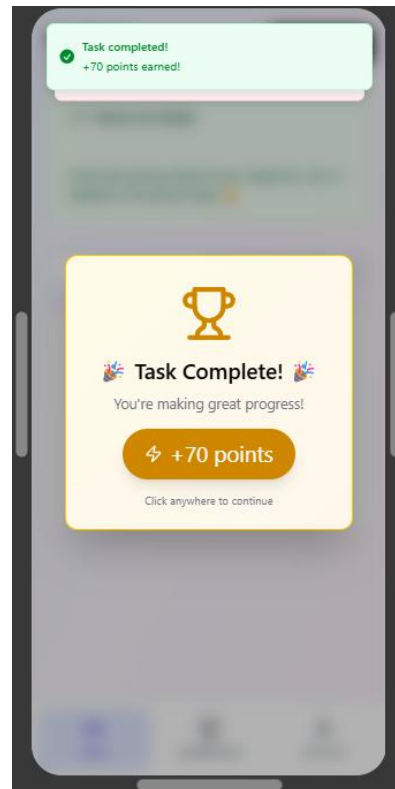


Figure 11: Leaderboard page showing rankings and scores of other users within the app

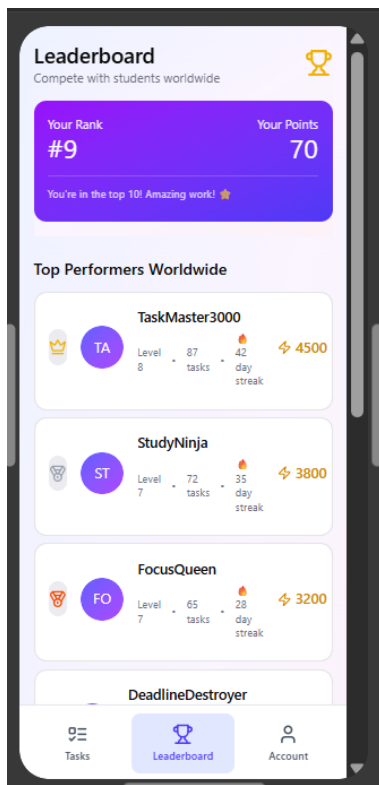
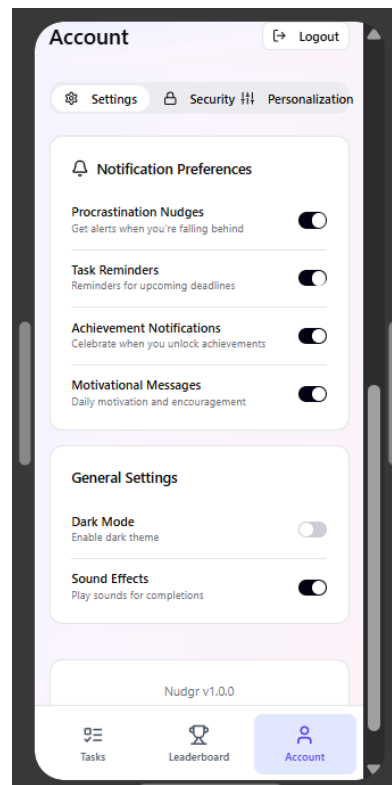


Figure 12: Account settings page where users can manage profile and app preferences



3.4.5 Phase 4: System Integration

During this phase, the system integration will connect the machine learning pipeline with the application development pipeline to create a single operational system. This ensures that predictive intelligence and gamified interventions work together smoothly in a real-time user environment.

3.4.5.1 Integration of Machine Learning Model

The trained machine learning model is set up as a service and accessed through an API. This is to allow real-time prediction requests from the application backend. This arrangement separates the prediction logic from the application layer. By doing so, it will enable the updates or retraining of the model this way without affecting the user interface.

The backend will preprocess inputs when a user submits a new questionnaire data using the same methods applied in model training. The processed data is then sent to the prediction API, which will return the procrastination risk in binary classification. This modular approach will improve maintainability and scalability while ensuring consistent predictions

The data flow between the application and the model is designed to be efficient and light. This prevents the prediction delays from harming the user experience. All prediction outputs will be logged for monitoring and potential future improvement of the model.

3.4.5.2 End-to-End System Workflow

The end-to-end workflow will start with user interaction and will end with adaptive gamified feedback. Once the system receives the user input, the system will produce a prediction regarding the risk of procrastination and checks if it exceeds a predetermined intervention threshold. The system will activate the feedback mechanisms when the prediction exceeds the threshold. The system will provide the feedback tailored to the identified risk factors.

Gamified responses will be dynamically chosen based on the prediction results and user history. The system may send motivational judges, task-focused challenges, or recovery mechanisms for streaks when the users are identified as at risk. In contrast, if users show consistent engagement, reinforcement feedback like badges or progress milestones will be provided to maintain positive behaviour.

The system will operate as both a predictive tool and an interactive platform for behavioural intervention through a closed feedback loop. It aims to enhance long-term engagement and self-regulation by continuously aligning prediction results with personalized gamified responses.

3.4.6 Tools and Techniques Used

To develop the proposed gamified procrastination intervention system, dual pipeline approach was used. The first pipeline is Machine Learning (ML) pipeline, and the second pipeline is the Application Development Pipeline.

The ML pipeline focuses on preparing the dataset, selecting features, training various machine learning models, and evaluating their performance using metrics. This pipeline will run on Google Colab platform, which offers an interactive Python environment that is suitable for data preprocessing, model training and evaluation. Not only that, Google Colab offers powerful computing resources, easy access to Python libraries, and datasets can be handled straightforwardly in different formats. Hence, this platform is suitable for this project which needs training, evaluation, and comparison of multiple machine learning models. By using Google Colab, this platform allows flexibility when experimenting with preprocessing methods, feature selection, and various predictive algorithms without worrying about the hardware constraints.

Meanwhile, the Application Development pipeline focuses on designing and building the mobile application prototype. This application will include gamification elements, a user interface, and backend logic. Flutter will be used to build the prototype that works on both Android and iOS. Flutter is chosen for this pipeline as it is helpful to attractive, responsive, and consistent interfaces across devices. This platform is also helpful when it comes to integrating gamified elements, features for task tracking, and real-time feedback mechanisms into one application. Interaction with APIs can also be done by Flutter as it is important to allow smooth interactions with the ML to predict procrastination risk.

By using Google Colab for ML pipeline and Flutter for application pipeline, this dual-pipeline approach ensure that both the predictive model and mobile application can be developed effectively. They can later be combined into a single system so it can support personalized, adaptive interventions for students.

4.0 Work Plan

This chapter outlines the work plan for the proposed gamified mobile application which is designed for the students to assist them in managing their procrastination. The work plan is divided into Capstone Project 1 and Capstone Project 2, detailing the activities planned, deliverables, milestones, associated risks, mitigation strategies, and estimated durations. This approach which is structured will ensure that the project progresses systematically from research and planning to implementation, evaluation, and final documentation.

4.1 Project Deliverables

Table 11 below summarizes the key deliverables for the overall project which covers both the research and system development outputs.

Table 11: Project deliverables with durations and scheduled weeks

No.	Deliverable	Duration
1.	Problem Definition and Project Scope Document	2 weeks (Week 1 – Week 2)
2.	Structured Literature Review Report	9 weeks (Week 2 – Week 10)
3.	Requirement Specification and Use Case Document	2 weeks (Week 9 – Week 10)
4.	Conceptual System Architecture and Process Flow Diagrams	2 weeks (Week 10 – Week 11)
5.	Machine Learning Pipeline Design Specification	2 weeks (Week 10 – Week 11)
6.	Pre-processed Dataset and Feature Definition File	2 weeks (Week 11 – Week 12)
7.	Trained Machine Learning Model for Procrastination Risk Prediction	3 weeks (Week 15 – Week 17)
8.	Model Evaluation and Performance Analysis Report	2 weeks (Week 17 – Week 18)
9.	Rule-Based Task Decomposition and Scheduling Logic	3 weeks (Week 17 – Week 19)
10.	Gamification Logic and Reward mechanism Design	2 weeks (Week 19 – Week 20)
11.	Backend Services and API Integration Layer	3 weeks (Week 17 – Week 19)
12.	Flutter-Based Mobile Application Frontend	2 weeks (Week 21 – Week 22)
13.	Integrated Gamified Procrastination Management System	3 weeks (Week 22 – Week 24)
14.	Usability Testing and Functional Validation Report	2 weeks (Week 26 – Week 27)
15.	Final Capstone Project Report and Documentation	2 weeks (Week 27 – Week 28)

The deliverables listed in the table above ensure that both theoretical foundations and technical components are sufficiently developed and documented. Documents that are research-based support the justification of choices for the design, whereas system-related deliverables demonstrate the practical implementation of the proposed solution.

4.2 Capstone Project 1 - Research and Planning Phase

The first phase which focuses on Capstone Project 1 emphasizes on problem understanding, research, requirement analysis, and system design. The activities in the phase establish a solid foundation so that it is easier to develop the prototype in Capstone Project 2.

4.2.1 Work Activities and Deliverables for Capstone Project 1

The activities listed in Table 12 below for Capstone Project 1 are sequenced to ensure that conceptual clarity is achieved before technical development begins. The literature review section helps to shape the methodology and system design. Hence, reducing the likelihood or rework in later stages.

Table 12: Capstone Project 1 work activities, deliverables, durations, risks, and mitigation strategies

Work Activity	Description	Deliverables	Duration	Risk Factors	Mitigation Strategy
Project Initiation and Scope Definition	Establish project context, define problem boundaries, state the objectives	Project scope and objectives document	2 weeks (Week 1 – Week 2)	Objectives misalign with the aim of the project	Early supervisor validation and iterative refinement
Preliminary Literature Exploration	Conduct exploratory reading to identify key themes in procrastination, ML, gamification, and task planning	Initial literature matrix and keyword map	2 weeks (Week 1 – Week 2)	Overly broad research focus	Narrow scope using inclusion and exclusion criteria
In-Depth Literature Review	Systematic review of empirical and theoretical studies related to procrastination, predictive modelling, gamification, and rule-based planning	Completed literature review chapter draft	7 weeks (Week 3 – Week 9)	Limited access to relevant or recent studies	Use multiple databases (MDPI, IEEE, Springer) and citation chaining
Problem Analysis and	Synthesize literature findings to identify limitations, gaps and	Problem statement and	2 weeks (Week 7 – Week 8)	Weak linkage between	Explicit mapping between gaps

Research Gap Identification	motivation for the proposed system	research gap justification		literature and problem	and project objectives
Requirement Elicitation and Analysis	Identify functional and nonfunctional requirements for ML prediction, and gamified feedback.	Requirement specification document	2 weeks (Week 9 - Week 10)	Ambiguous or incomplete requirements	Trace requirements back to literature and objectives
Dataset Identification and Feasibility Analysis	Identify suitable datasets or data sources for procrastination prediction and assess feasibility	Dataset description and justification section	5 weeks (Week 2 – Week 6)	Dataset mismatch with objectives	Use benchmark datasets or simulated data when necessary
Machine Learning Pipeline Design	Design data preprocessing, feature selection, model training, and evaluation strategy	ML pipeline design specification	2 weeks (Week 10 – Week 11)	Overly complex impractical pipeline	Strat with baseline models and justify incremental complexity
System Architecture and Workflow Design	Explain overall system architecture, frontend-backend interaction, and ML integration workflow	Architecture diagrams and process flowcharts	2 weeks (Week 10 – Week 11)	Poor modularity or tight coupling	Segment the explanations into subsections for clearer understanding
Methodology Consolidation	Integrate ML, rule-based AI, and application development into a coherent methodology	Finalized methodology chapter draft	3 weeks (Week 10 – Week 12)	Inconsistencies across sections	Cross-check methodology against objective
Documentation Review and Refinement	Revise all CP 1 documents for clarity, coherence, and academic rigor	Final CP 1 report draft	2 weeks (Week 13 – Week 14)	Incomplete or inconsistent documentation	Structured proofreading and supervisor feedback

4.3 Capstone Project 2: Implementation and Evaluation Phase

This phase focuses on the development of the system, implementation of the machine learning, integration of the application, testing and evaluation. This phase transforms the planned design done in Capstone Project 1 into a functional system.

4.3.1 Work Activities and Deliverables for Capstone Project 2

Table 13 below highlights the development and evaluation that will be done iteratively to ensure reliability and usability of the system. The integration of rule-based task decomposition with predictive modelling supports personalized and adaptive procrastination management.

Table 13: Capstone Project 2 work activities, deliverables, durations, risks, and mitigation strategies

Work Activity	Description	Deliverables	Duration	Risk Factor	Mitigation Strategy
Data Preparation	Preprocess dataset for ML training	Cleaned dataset	2 weeks (Week 15 – Week 16)	Poor data quality	Apply data cleaning and validation
ML Model Development	Train and tune procrastination prediction model	Trained ML model	3 weeks (Week 15 – Week 17)	Overfitting or low accuracy	Use validation and tuning techniques
Rule-Based Task Decomposition	Implement AI rules to break task into subtasks	Rule-based module	3 weeks (Week 17 – Week 19)	Ineffective task breakdown	Iterative rule refinement
Application Development	Develop mobile application with gamification features	Functional mobile app prototype	4 weeks (Week 19 – Week 22)	Integration complexity	Develop in modular stages
System Integration	Integrate ML model and rule-based logic into app	Integrated system	3 weeks (Week 22 – Week 24)	Compatibility issues	Test components independently
Testing and Evaluation	Conduct functional and usability testing	Testing and evaluation report	2 weeks (Week 26 – Week 27)	Undetected system bugs	Perform iterative testing
Final Documentation	Prepare final report and presentation	Final report slides	2 weeks (Week 27 – Week 28)	Time constraints	Prioritize critical sections

4.4 Project Milestones with Risks and Mitigation

Table 14 highlights the key projects milestones along with its associated risks and mitigation strategies. By identifying the risks at each milestone allows for a proactive mitigation, thus reducing delays and improving overall project quality.

Table 14: Key project milestones with associated risks and mitigation strategies for proactive management

Milestone	Risk	Mitigation Strategy
Completion of Literature Review	Insufficient coverage of relevant studies	Expand search keywords, use multiple databases, supervisor review
Finalization of Problem Definition and Scope	Scope creep or unclear project boundaries	Formal scope validation and milestone sign-off
Finalization of Methodology	Inappropriate model or rule-based approach selection	Validate design choices using prior studies and expert feedback
Completion of System and Rule-Based Design	Design infeasibility during implementation	Conduct design walkthroughs and modular validation
ML Model Training and Validation	Poor prediction accuracy or overfitting	Hyperparameter tuning and cross-validation
Rule-Based Task Decomposition Implementation	Ineffective or rigid task breakdown rules	Iterative rule refinement and scenario-based testing
Application Integration	System incompatibility across components	Modular testing and phased integration
System Testing and Evaluation	Incomplete performance or usability validation	Structured testing protocols and metric-based evaluation
Final Submission	Incomplete or inconsistent documentation	Early drafting, iterative reviews, supervisor feedback

4.5 Gantt Chart

The project timeline for both the phases which are Capstone Project 1 and Capstone Project 2 is illustrated using Gantt charts.

Figure 13 is the Gantt chart for Capstone Project 1 which highlights the planning, literature review, methodology, and system design activities. Figure 14 is the Gantt chart for Capstone Project 2 that focuses on the implementation, testing, and documentation phases.

These charts provide a visual overview of the durations of tasks, dependencies, and milestones to track the project effectively.

Figure 13: Gantt Chart for Capstone Project 1

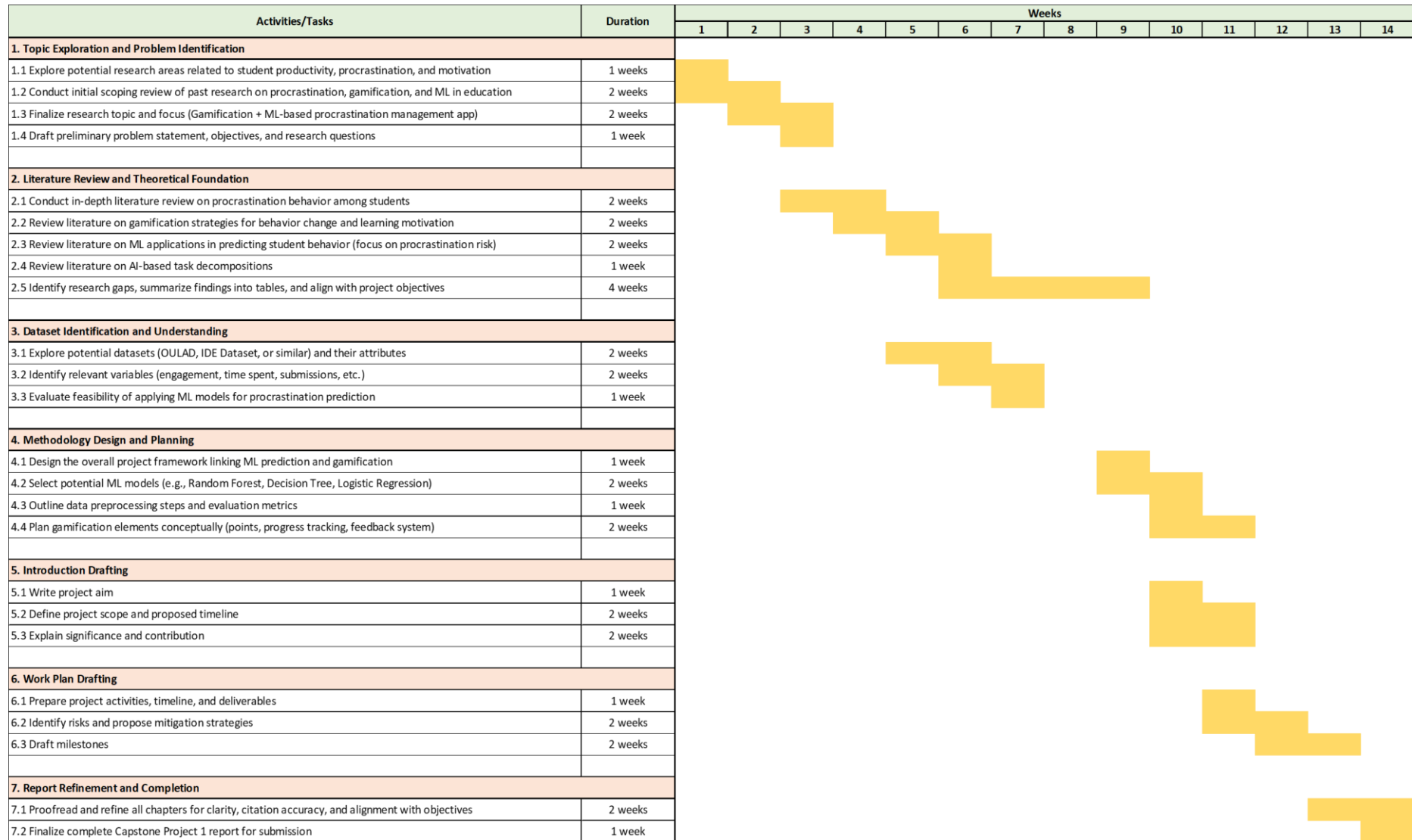
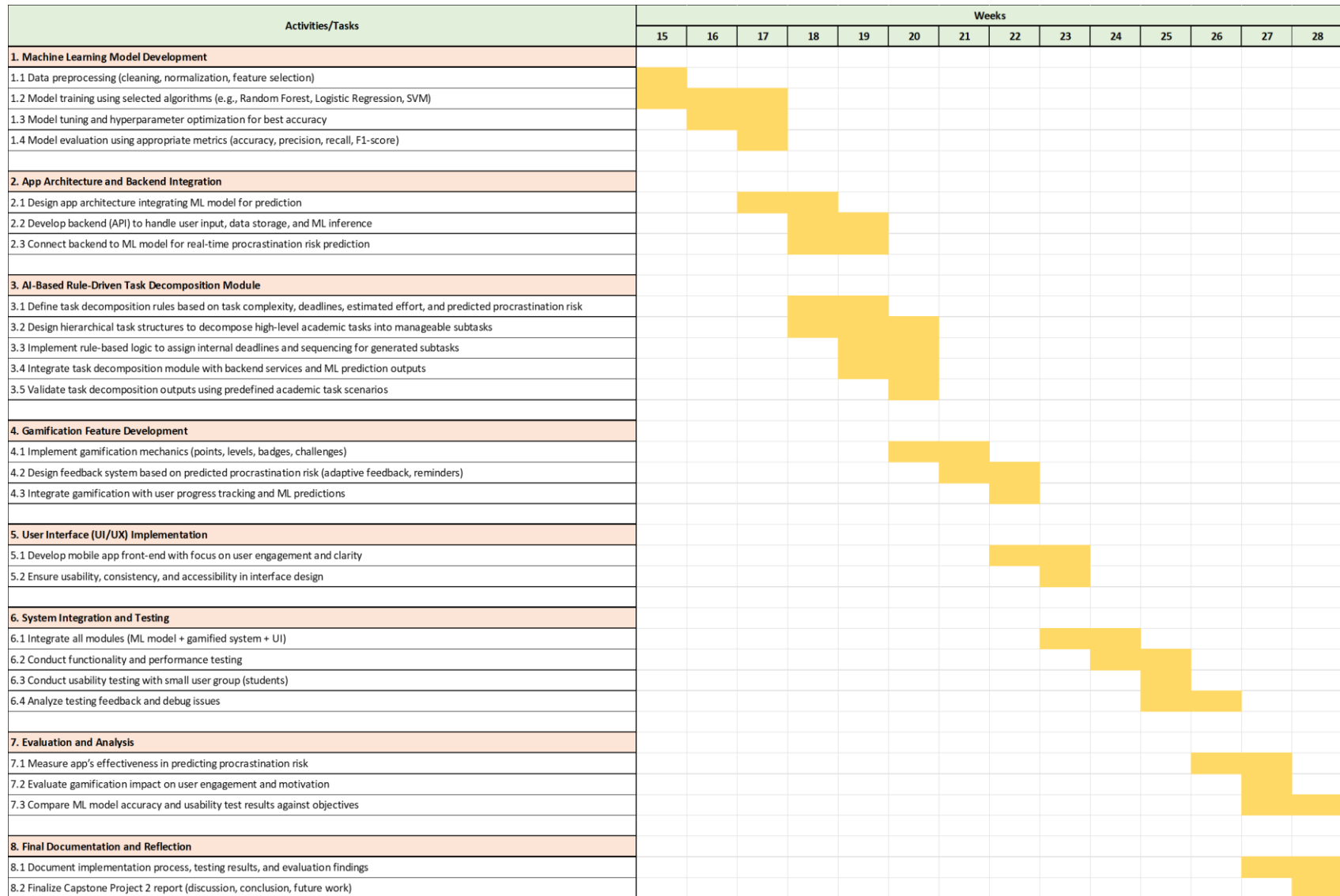


Figure 14: Gantt Chart for Capstone Project 2



4.6 Chapter Summary

This chapter has presented a structured work plan for the proposed project. It includes detailing activities, deliverables, risks, mitigation strategies and timelines across both the capstone phases. By following this structured approach, it is ensured that the project can progress logically from research and planning to implementation and assessment. This supports the successful development of the gamified procrastination management application.

5.0 References

- Abidi, S. M. R., Zhang, W., Haidery, S. A., Rizvi, S. S., Riaz, R., Ding, H., & Kwon, S. J. (2020). Educational Sustainability through Big Data Assimilation to Quantify Academic Procrastination Using Ensemble Classifiers. *Sustainability*, 12(15), 6074. <https://www.mdpi.com/2071-1050/12/15/6074>
- Ahmed, I., Bernhardt, G. V., & Shivappa, P. (2023). Prevalence of Academic Procrastination and Its Negative Impact on Students. *Biomedical and Biotechnology Research Journal (BBRJ)*, 7(3), 363-370. https://doi.org/10.4103/bbrj.bbrj_64_23
- Almalawi, A., Soh, B., Li, A., & Samra, H. (2024). Predictive Models for Educational Purposes: A Systematic Review. *Big Data and Cognitive Computing*, 8(12), 187. <https://www.mdpi.com/2504-2289/8/12/187>
- Amit, A. J., Gautam Shankararam, S. R., Pradeep, P., Perumalraja, R., & Kamalesh, S. (2021). *Framework For Preventing Procrastination And Increasing Productivity*
- Baylin, S. M., & Lomibao, L. S. (2024). Modeling Students' Procrastination Using Gaussian - Bernoulli Mixed Naïve Bayes Method. *Journal of Innovations in Teaching and Learning*, 4(1), 7-12. <https://pubs.sciepub.com/jitl/4/1/2>
- Chen, X., Wu, M., Dong, G., Cui, L., Qu, B., & Zhu, Y. (2025). Social support and academic procrastination in health professions students: the serial mediating effect of intrinsic learning motivation and academic self-efficacy. *Advances in Health Sciences Education*, 30(4), 1143-1160. <https://doi.org/10.1007/s10459-024-10394-4>
- Cheng, S.-L., Chang, J.-C., Quilantan-Garza, K., & Gutierrez, M. L. (2023). Conscientiousness, prior experience, achievement emotions and academic procrastination in online learning environments. *British Journal of Educational Technology*, 54(4), 898-923. <https://doi.org/https://doi.org/10.1111/bjet.13302>
- Elkawagy, M., Elgendy, I.-A., Muthanna, A., Alkanhel, R.-I., & Elbeh, H. (2025). Enhancing Hierarchical Task Network Planning through Ant Colony Optimization in Refinement Process. *Computers, Materials & Continua*, 84(1), 393--415. <http://www.techscience.com/cmc/v84n1/61748>
- Fentaw, Y., Moges, B. T., & Ismail, S. M. (2022). Academic Procrastination Behavior among Public University Students. *Education Research International*, 2022(1), 1277866. <https://doi.org/https://doi.org/10.1155/2022/1277866>
- García-Ros, R., Pérez-González, F., Tomás, J. M., & Sancho, P. (2023). Effects of self-regulated learning and procrastination on academic stress, subjective well-being, and academic

- achievement in secondary education. *Current Psychology*, 42(30), 26602-26616.
<https://doi.org/10.1007/s12144-022-03759-8>
- Hebebe, M. T., & Alan, S. (2021). Gamification in education: An overview of the literature. *Current studies in social sciences*, 174.
- Huang, H., Ding, Y., Liang, Y., Wan, X., Peng, Q., Zhang, Y., Kong, D., & Chen, C. (2023). The association between perfectionism and academic procrastination among undergraduate nursing students: The role of self-efficacy and resilience. *Nursing Open*, 10(10), 6758-6768. <https://doi.org/10.1002/nop2.1922>
- Jamil, L., Ashouri, A., Zamirinejad, S., & Mahaki, B. (2021). Investigating the structural model of procrastination based on transdiagnostic factors. *Med J Islam Repub Iran*, 35, 120.
<https://doi.org/10.47176/mjiri.35.120>
- Jing, H., Sharif, S., & Chua, B. S. (2025). Negative Perfectionism and Academic Procrastination: How Self-Efficacy Shapes the Link in Chinese University Students. *Malaysian Journal of Social Sciences and Humanities (MJSSH)*, 10(6), e003440.
- Kienle, C., Alt, B., Arenz, O., & Peters, J. (2025). LODGE: Joint Hierarchical Task Planning and Learning of Domain Models with Grounded Execution. *arXiv preprint arXiv:2505.13497*.
- Kirchner-Krath, J., Schmidt-Kraepelin, M., Schmähl, K., Schütz, C., Morschheuser, B., & Sunyaev, A. (2025). Behavior Change Support Systems for Self-Treating Procrastination: Systematic Search in App Stores and Analysis of Motivational Design Archetypes [Original Paper]. *J Med Internet Res*, 27, e65214.
<https://doi.org/10.2196/65214>
- Kufel, J., Bargieł-Łączek, K., Kocot, S., Koźlik, M., Bartnikowska, W., Janik, M., Czogalik, Ł., Dudek, P., Magiera, M., Lis, A., Paszkiewicz, I., Nawrat, Z., Cebula, M., & Gruszczyńska, K. (2023). What Is Machine Learning, Artificial Neural Networks and Deep Learning?—Examples of Practical Applications in Medicine. *Diagnostics*, 13(15), 2582. <https://www.mdpi.com/2075-4418/13/15/2582>
- Memon, M. M., Hashmani, M. A., Jameel, S. M., Junejo, S., & Raza, K. (2020). Analysis of student procrastinatory behavior in virtual learning environments using machine learning. *Journal of Hunan University Natural Sciences*, 47(10).
- Mohd Talib, N. I., Abd Majid, N. A., & Sahran, S. (2023). Identification of Student Behavioral Patterns in Higher Education Using K-Means Clustering and Support Vector Machine. *Applied Sciences*, 13(5), 3267. <https://www.mdpi.com/2076-3417/13/5/3267>

- Noviana Khoirunnisa, R., Soetjipto, B. E., Chusniyah, T., Rahmawati, H., & Boonroungrut, C. (2024). Academic Self-Efficacy and Future Time Perspective Predict Academic Procrastination: The Mediating Function of Academic Self-Control: Predict academic procrastination. *International Journal of Body, Mind and Culture*, 11(5), 669-686. <https://doi.org/10.22122/ijbmc.v11i5.806>
- Öztaş, G., & Akçapınar, G. (2025). Predicting students' academic procrastination tendencies using online learning trajectories. *Educational Technology & Society*, 28, 77-93. [https://doi.org/10.30191/ETS.202504_28\(2\).RP05](https://doi.org/10.30191/ETS.202504_28(2).RP05)
- Pengel, N., Martin, A., Meissner, R., Arndt, T., Neumann, A. T., de Lange, P., & Wollersheim, H.-W. (2021). TecCoBot: Technology-aided support for self-regulated learning. *arXiv preprint arXiv:2111.11881*.
- Pereira, M. B., Classe, T. M. d., & Siqueira, S. W. M. (2025). Investigating Gamification to Reduce Procrastination - Systematic Literature Review. *Journal on Interactive Systems*, 16(1), 302-319. <https://doi.org/10.5753/jis.2025.5412>
- Prasad, A., Koller, A., Hartmann, M., Clark, P., Sabharwal, A., Bansal, M., & Khot, T. (2024). Adapt: As-needed decomposition and planning with language models. Findings of the Association for Computational Linguistics: NAACL 2024,
- Rusdi, M., Hidayah, N., Rahmawati, H., & Hitipeuw, I. (2020). Academic procrastination data of students in Makassar, Indonesia. *Data in Brief*, 33, 106608. <https://doi.org/https://doi.org/10.1016/j.dib.2020.106608>
- Sartana, B. T., Nugroho, S. M. S., Yuhana, U. L., & Purnomo, M. H. (2025). Multimodel Prediction Score Based on Academic Procrastination Behavior in E-Learning. *Jurnal Nasional Pendidikan Teknik Informatika: JANAPATI*, 14(1), 1-12. <https://doi.org/10.23887/janapati.v14i1.85880>
- Shen, X., & Yuan, C. (2021). A College Student Behavior Analysis and Management Method Based on Machine Learning Technology. *Wireless Communications and Mobile Computing*, 2021(1), 3126347. <https://doi.org/https://doi.org/10.1155/2021/3126347>
- Sirois, F. M. (2023). Procrastination and Stress: A Conceptual Review of Why Context Matters. *International Journal of Environmental Research and Public Health*, 20(6), 5031. <https://www.mdpi.com/1660-4601/20/6/5031>
- Staller, M. S., & Koerner, S. (2021). Beyond Classical Definition: The Non-definition of Gamification. *SN Computer Science*, 2(2), 88. <https://doi.org/10.1007/s42979-021-00472-4>

- Sun, H. (2024). Effect of gamification on persuasive application on academic procrastination : A case study of Forest: Focus for Productivity. [40], 45 , school = , School of Informatics.
- Yamani, H. A. (2021). A Conceptual Framework for Integrating Gamification in eLearning Systems Based on Instructional Design Model. *International Journal of Emerging Technologies in Learning (iJET)*, 16(04), pp. 14-33. <https://doi.org/10.3991/ijet.v16i04.15693>
- Yan, B., & Zhang, X. (2022). What Research Has Been Conducted on Procrastination? Evidence From a Systematical Bibliometric Analysis [Systematic Review]. *Frontiers in Psychology, Volume 13 - 2022*. <https://doi.org/10.3389/fpsyg.2022.809044>
- Yao, M., Sahebi, S., & Feyzi Behnagh, R. (2020). Analyzing student procrastination in MOOCs: a multivariate Hawkes approach. Proceedings of the 13th conference on educational data mining (EDM2020),
- Yuan, J., Qiu, X., Wu, J., Guo, J., Li, W., & Wang, Y.-G. (2024). Integrating behavior analysis with machine learning to predict online learning performance: A scientometric review and empirical study. *arXiv preprint arXiv:2406.11847*.